Machine-Learning and Operational Subseasonal Forecasts over India: A Two-Season Benchmark across Winter and Monsoon Regimes

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Abstract

Subseasonal-to-seasonal (S2S) forecasts at two-to-six-week leads support Indian agriculture and water management, yet the skill of new machine-learning (ML) S2S systems over India remains sparsely documented. We present two contrasting Indian case studies with a common verification framework: a synoptically driven winter case (JFM 2026, 90 daily initializations), where Spire AI-S2S is evaluated alongside FuXi-S2S, DLESyM, ECMWF, UKMO, and NCEP, and a convectively dominated monsoon case (JJAS 2019, 35 common initializations) for ECMWF, UKMO, NCEP, and FuXi-S2S. Forecasts are verified over Indian land against ERA5 and IMD rainfall using anomaly correlation (ACC), RMSE, bias, CRPSS, and spread–skill ratio. In JFM 2026, Spire AI-S2S has the highest precipitation ACC at every lead (week-1 ACC 0.78 versus 0.73 for ECMWF and UKMO); a paired bootstrap shows this lead is significant over every precipitation-capable system through week 3 and over ECMWF at every lead. Spire also degrades most gracefully for Z500, significantly exceeding FuXi-S2S and DLESyM at all leads, with the gap widening after week 1; both cross into negative ACC by week 4. In JJAS 2019, by contrast, precipitation skill collapses rapidly for all available systems: every individual-model ACC is at most 0.04 by week 3, where none differs significantly from zero. The contrast points to a physical predictability difference rather than a model leaderboard: winter rainfall is tightly coupled to organized large-scale circulation and western disturbances, whereas monsoon rainfall depends on multiscale convection and active–break variability that are harder to predict at weekly leads. Deterministic and probabilistic rankings, moreover, need not agree. We provide an early, truth-aware benchmark of AI and operational S2S guidance over India while making explicit the limits of two seasons and unequal model availability.

Keywords: subseasonal-to-seasonal prediction; machine learning; precipitation; India; monsoon; ensemble verification

1 Introduction

Subseasonal-to-seasonal (S2S) prediction, spanning lead times from roughly two weeks to two months, bridges medium-range weather forecasting and seasonal climate outlooks (Vitart et al., 2017; National Academies, 2016). Forecasts at these leads are increasingly demanded for agricultural planning, reservoir operations, energy load management, and disaster preparedness (White et al., 2017), yet they remain among the most difficult products to produce skillfully: the atmosphere has largely lost its initial-condition memory, while slowly varying boundary forcings have not yet come to dominate (Mariotti et al., 2020). Over India, skillful multi-week outlooks carry particular weight. Winter precipitation over northern India and the Western Himalaya, produced chiefly by western disturbances (WDs)—extratropical synoptic systems embedded in the subtropical westerly jet (Dimri et al., 2015; Hunt et al., 2018)—sustains the rabi (winter) cropping season and builds the Himalayan snowpack that feeds the Indus and Ganges during the subsequent melt season. The summer monsoon (June through September), by contrast, supplies roughly three-quarters of India's annual rainfall through transient convective systems organized by intraseasonal oscillations (Zhang, 2005; Wheeler and Hendon, 2004), and its subseasonal predictability is a long-standing operational priority for the India Meteorological Department (IMD) and the Ministry of Earth Sciences (Rao et al., 2019; Pattanaik et al., 2019). These two seasons are governed by different dynamics and, as we show below, by different limits on predictability.

In parallel, weather prediction is being reshaped by machine-learning (ML) forecast systems trained on reanalysis data. Deterministic data-driven models such as Pangu-Weather and GraphCast now match or exceed operational numerical weather prediction (NWP) on standard medium-range scores (Pathak et al., 2022; Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024; Rasp et al., 2024), probabilistic and hybrid successors have followed (Price et al., 2025; Kochkov et al., 2024), and operational centers have begun running data-driven systems alongside their NWP suites (Lang et al., 2024). Extension of these methods to the subseasonal range is more recent: FuXi-S2S, a transformer-based system producing 42-day ensemble forecasts, has been reported to outperform conventional S2S models on several global metrics (Chen et al., 2024); diffusion-based Earth-system emulators such as DLESyM (Cresswell-Clay et al., 2025) have followed; and commercial providers now issue AI-based S2S products (Spire Global, 2025). Published evaluations of these systems have, however, emphasized global or large-scale diagnostics, while regional precipitation skill over densely populated, hydrologically sensitive domains such as India remains largely undocumented—and essentially no published work compares more than one ML S2S system against operational dynamical baselines over the same domain and the same verification framework.

A further gap concerns the evaluation setting itself. Existing ML weather-forecast evaluations are typically run on a single season or a short real-time window, which cannot distinguish a genuinely better forecast system from one that merely performs well in an easy predictability regime. India offers an unusually clean natural test of this distinction: its two principal rainy seasons are governed by essentially different physics. Winter precipitation is organized by synoptic-scale WDs with several days of predictable structure, while monsoon precipitation is dominated by smaller-scale convection embedded in a noisier large-scale envelope, historically associated with a shorter predictability horizon (de Andrade et al., 2019; Pegion et al., 2019). A forecast system that performs well only in the synoptically organized winter season may not be skillful in general; a system that holds skill in the monsoon, where the signal-to-noise ratio is lower, is making a stronger claim.

We exploit this contrast through two complementary, deliberately asymmetric case studies rather than through a single homogeneous hindcast archive. The JFM 2026 case is a daily initialized real-time winter benchmark that includes Spire AI-S2S, making it the main novelty of this study. The JJAS 2019 case is a monsoon benchmark on common operational initialization dates for ECMWF, UKMO, NCEP, and FuXi-S2S, with DLESyM used only where its available forecasts support a matched sensitivity comparison. Both cases use the same India domain, weekly-lead definitions, land mask, climatology construction, and verification metrics, but the available model set is not identical across seasons.

Specifically, we address four questions:

1. How does weekly precipitation and Z500 skill decay from week 1 to week 6 over India in a winter circulation regime and a monsoon rainfall regime?
2. Does Spire AI-S2S add value in the JFM 2026 winter case relative to FuXi-S2S and operational dynamical ensembles?
3. In the JJAS 2019 monsoon case, do ECMWF, UKMO, NCEP, and FuXi-S2S retain usable rainfall skill beyond the first two forecast weeks?
4. Does deterministic skill translate into probabilistic skill and calibrated spread?

We verify ensemble-mean (or Spire mean) forecasts against ERA5 and IMD gridded rainfall using the cosine-weighted anomaly correlation coefficient (ACC), root-mean-square error (RMSE), and bias, complemented by a common-framework probabilistic evaluation (CRPSS, spread–skill ratio). The remainder of the paper is organized as follows. Section 2 describes the verification data and study domain, Section 3 the six forecast systems, and Section 4 the verification methodology. Results are presented in Section 5 and discussed in Section 6; limitations and conclusions follow in Sections 7–8.

2 Data and Study Domain

2.1 Study domain and regions

The study domain covers the Indian region, 5°–38°N, 65°–100°E, with all aggregate skill metrics computed over *land points only*. Land points are defined as the union of the four India Meteorological Department (IMD) homogeneous rainfall regions—Northwest, Central, South Peninsula, and East/Northeast India—following the state grouping of Pai et al. (2014); the region masks are constructed by assigning each 1.5° grid point to the region whose constituent Survey-of-India state polygons contain it, so land/ocean separation is a byproduct of the state-boundary geometry rather than a separate coastline product. Ocean grid cells (Arabian Sea, Bay of Bengal) are therefore automatically excluded from every area average, since they would otherwise dominate the domain mean and distort precipitation skill. Spatial means are weighted by the cosine of latitude throughout (Section 4.3). The four regions are mutually exclusive and their union is the "All India" land mask used for every domain-aggregated score in this paper (Figure 1).

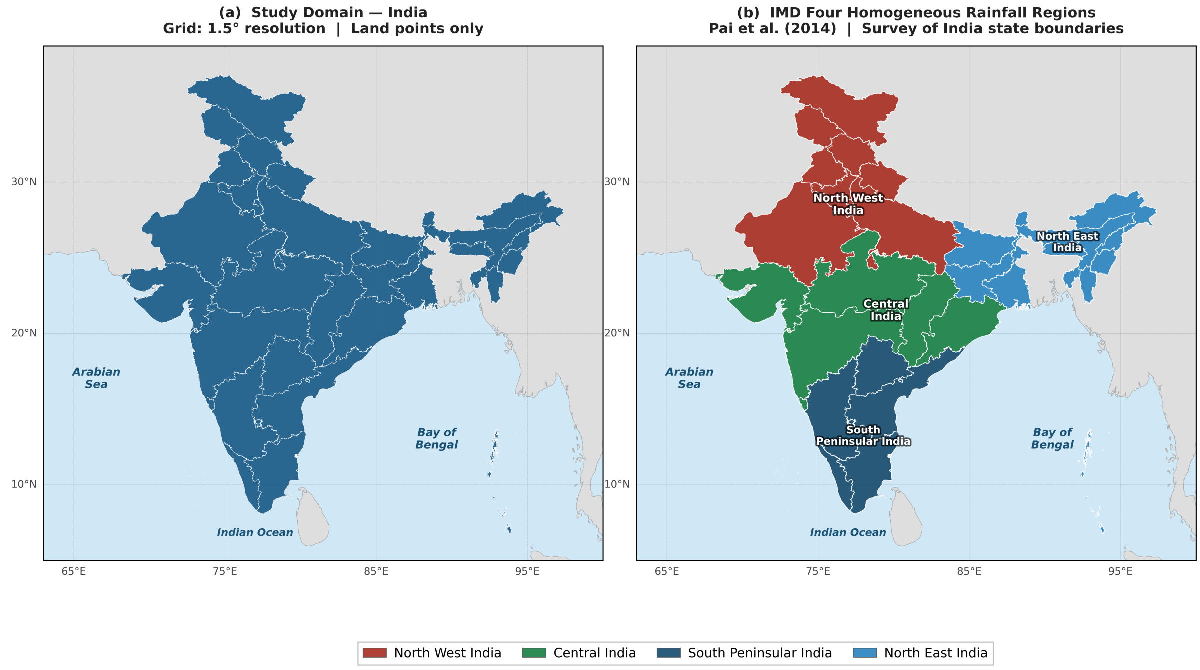

Figure 1: (a) The India study domain (land points only, common 1.5° verification grid). (b) The four IMD homogeneous rainfall regions (Northwest, Central, South Peninsula, East/Northeast India), delineated by Survey-of-India state boundaries following Pai et al. (2014). All aggregate skill metrics in this paper are computed over land points only, using cosine-latitude area weights.

2.2 Verification references

Three observational references are used, chosen to match the best available truth in each season and to avoid conflating model skill with reference-data artifacts.

- **Precipitation, JFM 2026:** hourly total precipitation from the Analysis-Ready, Cloud-Optimized ERA5 archive (ARCO-ERA5; Carver and Merose, 2023), summed to true 24-hour daily totals (00–00 UTC). The 2026 IMD gridded rainfall product is not yet available for this season, so ERA5 is the only complete reference; we discuss the implications of treating a reanalysis as truth in Section 7.
- **Precipitation, JJAS 2019:** the IMD 0.25° daily gauge-based gridded rainfall product (Pai et al., 2014), constructed from roughly 3000 quality-controlled rain-gauge stations by ordinary kriging. Because this reference is independent of any reanalysis, it removes the possibility that an ERA5-trained ML system is rewarded simply for reproducing ERA5's own precipitation biases.
- **Z500, both seasons:** ERA5 (Hersbach et al., 2020). For JFM 2026, Z500 truth is taken from ERA5 CDS daily files; for JJAS 2019 we use the local WeatherBench2 ERA5 archive (Rasp et al., 2024), which provides matching geopotential fields at 1.0° resolution for the 2019 period.

Because the JFM and JJAS precipitation truths differ (ERA5 versus IMD), all within-season model rankings are valid, but the absolute precipitation skill level should not be compared directly across seasons; we return to this point in Section 7.

Table 1: Forecast systems evaluated. "Native res." is each model's own horizontal grid; all are regridded to the common 1.5° verification grid before scoring. CF = control, PF = perturbed member. Members are the actual archived ensemble size used for scoring (Section 7); ECMWF and FuXi-S2S differ between seasons.

System	Type	Native res.	Members	Horizon	Seasons
Spire AI-S2S	ML (data-driven)	0.5°	mean+s.d.	46 d	JFM 2026
FuXi-S2S	ML (transformer)	1.5°	11 / 50*	42 d	Both
DLESyM	ML (diffusion)	0.25°	80	48 d	JFM; JJAS sens.
ECMWF	Dynamical, coupled	Tco319 (∼36 km)	101 / 51*	46 d	Both
UKMO GloSea6	Dynamical, coupled	N216 (∼60 km)	4	60 d	Both
NCEP CFSv2	Dynamical, coupled	T126 (∼100 km)	16	44 d	Both

*JFM 2026 / JJAS 2019 (archived ensemble size differs by season).

2.3 Forecast systems

We evaluate six S2S forecast systems, summarized in Table 1: three machine-learning systems (Spire AI-S2S, FuXi-S2S, DLESyM) and three operational dynamical systems (ECMWF extended range, UKMO GloSea6, NCEP CFSv2). The comparison is intentionally not symmetric across seasons. All six systems are used in the JFM 2026 winter case where Spire is available. The main JJAS 2019 monsoon case uses the 35 common dates shared by ECMWF, UKMO, NCEP, and FuXi-S2S; Spire is unavailable for 2019, and DLESyM is retained only for a smaller sensitivity comparison because its usable monsoon initialization set is more limited and it has no precipitation channel. Full system descriptions, including how each variable is delivered and any unit conversions applied before scoring, are given in Section 3.

3 Forecast Systems

We evaluate six S2S forecast systems spanning three machine-learning systems and three dynamical NWP ensembles. The dynamical systems (ECMWF, UKMO, NCEP) are obtained from the WWRP/WCRP S2S database (Vitart et al., 2017); the ML systems were received directly from their respective providers. All six systems are evaluated over JFM 2026. In JJAS 2019, the main precipitation and Z500 benchmark is restricted to ECMWF, UKMO, NCEP, and FuXi-S2S on 35 common initializations; DLESyM is evaluated separately where its available fields permit a smaller matched comparison.

3.1 Spire AI-S2S

Spire AI-S2S (Spire Global, 2025) is a commercially produced, ML-based subseasonal forecasting system developed by Spire Global. Forecasts are delivered on a 0.5° global grid at daily temporal resolution to a 46-day horizon, initialized from 00 UTC ERA5 analyses. Rather than distributing individual ensemble members, the system supplies ensemble summary products—the mean and standard deviation, used here for both deterministic and probabilistic verification. Spire delivers daily total precipitation (mm day^{-1}) directly, so no unit conversion is required. Forecasts were received for the JFM 2026 season under a research data-use agreement; 2019 data were not available, so Spire is excluded from the JJAS 2019 comparison.

3.2 FuXi-S2S

FuXi-S2S (Chen et al., 2024) is a transformer-based ML subseasonal forecasting system trained on ERA5 reanalysis. The published system generates 42-day global forecasts on a 1.5° grid, initialized from 00 UTC ERA5 fields; we use the ensemble mean for deterministic scores and the member spread for probabilistic verification. **The archived ensemble size differs between the two seasons evaluated here**: 11 members (1 control, 10 perturbed) for JFM 2026, versus 50 members for JJAS 2019. This reflects what was available in the archive obtained for each season rather than a deliberate experimental design choice, and it means the JJAS 2019 FuXi-S2S probabilistic scores (CRPS, spread–skill ratio) are built from a substantially larger ensemble than the JFM 2026 scores; we flag this explicitly in Section 7 rather than treat FuXi-S2S as a single fixed-configuration system across seasons. The FuXi-S2S precipitation channel is archived as a mean rate ($\mathrm{mm\,h^{-1}}$) rather than a daily total and is multiplied by 24 before scoring to convert to $\mathrm{mm\,day^{-1}}$ (Section 4.3).

3.3 DLESyM

DLESyM (Deep Learning Earth System Model; Cresswell-Clay et al. 2025) is a diffusion-based coupled atmosphere–land–ocean ML system trained on ERA5 and CMIP6 simulations. Forecasts run on a 0.25° global grid to a 48-day horizon and are delivered as an 80-member ensemble—the largest ensemble in this comparison—which is particularly advantageous for probabilistic verification. DLESyM outputs 6-hourly fields, which we average to daily means before aggregating to weekly lead windows; geopotential is converted to geopotential height by dividing by $g = 9.80665\,\mathrm{m\,s^{-2}}$. DLESyM does not currently provide a precipitation channel and is therefore evaluated for Z500 only in this study. Three per-initialization forecast stores were found to be corrupted (one reading as all-zero, two missing entirely) and are excluded from the comparable-initialization sets; an automated guard rejects any store that reads as all-zero at the first lead.

3.4 ECMWF extended-range ensemble

The ECMWF Integrated Forecasting System (IFS) extended-range ensemble (Vitart, 2014) is a fully coupled atmosphere–ocean–land–sea-ice model run twice weekly to day 46. We use the 00 UTC runs archived on the S2S database (Vitart et al., 2017) at 1.5° (interpolated from the native Tco319, ≈36 km, spectral grid); we use the ensemble mean for deterministic scores and the full ensemble for probabilistic verification. **As with FuXi-S2S, the archived ensemble size differs between seasons**: 101 members (1 control, 100 perturbed) for JFM 2026 versus 51 members for JJAS 2019, again reflecting archive availability rather than a deliberate design choice (Section 7). ECMWF archives total precipitation as a running cumulative accumulation from initialization; the weekly-mean rate is recovered by differencing the cumulative totals at the start and end of each lead window and dividing by the number of days (Section 4.3). Geopotential height is delivered directly and requires no unit conversion.

3.5 UKMO GloSea6

The UK Met Office Global Seasonal Forecasting System version 6 (GloSea6; MacLachlan et al., 2015) is a fully coupled atmosphere–ocean–land system run on the Met Office Unified Model (MetUM) at ∼60 km (N216) resolution to a 60-day horizon. We use the 00 UTC runs archived on the S2S database at 1.5°, comprising one control forecast plus three perturbed members (a four-member ensemble), which limits UKMO's probabilistic scores relative to the larger ensembles but still supports full deterministic verification. UKMO delivers total precipitation as a cumulative accumulation, processed identically to ECMWF, and geopotential height already in gpm (no unit conversion).

3.6 NCEP CFSv2

The NCEP Climate Forecast System version 2 (Saha et al., 2014) is a fully coupled atmosphere–ocean–land system, operational since 2011, resolved at T126 (≈100 km) with 64 hybrid sigma-pressure levels. Forecasts are archived on the S2S database at 1.5° to a 44-day horizon; we use the archived 16-member ensemble (1 control, 15 perturbed) from each 00 UTC initialization, stable across both seasons evaluated here. Precipitation is archived as a cumulative accumulation and de-accumulated identically to ECMWF and UKMO.

4 Verification Framework

4.1 Weekly lead windows and aggregation

Each forecast is partitioned into six nonoverlapping weekly lead windows—week 1 to forecast days 1–7, week 2 to days 8–14, and so on through week 6 (days 36–42), where day 1 denotes the first 24 h after initialization. For each lead window, daily fields are averaged to a weekly-mean field; a weekly mean is scored only if at least two daily values fall inside the window (otherwise the model/init/week cell is dropped and logged), so a small number of weekly means in the published tables may be built from fewer than the full seven days where a forecast archive has isolated missing days, rather than from a complete week. All forecasts and the observational references are remapped to a common 1.5° latitude–longitude grid (matching the coarsest delivered system) prior to verification, so that no high-resolution system is penalized for representing fine scales the others cannot. Spatial means use $\cos\varphi$ area weights and are restricted to Indian land points (Section 2.1).

Because the precipitation-capable systems deliver rainfall in different native forms, each is reconciled to a common weekly-mean rate (mm day^{-1}) before scoring. Spire and FuXi-S2S (after the ×24 unit conversion, Section 3.2) deliver daily values that are simply averaged over the seven days of the lead window. ECMWF, UKMO, and NCEP archive precipitation as a running cumulative accumulation from initialization; the weekly-mean rate is recovered by differencing the cumulative totals at the last day (d_e) and the day before the window opens ($d_s - 1$), then dividing by the number of days N:

$$\overline{P}_w = \frac{\mathrm{TP}(d_e) - \mathrm{TP}(d_s - 1)}{N}, \tag{1}$$

with $\mathrm{TP}(d_s - 1) = 0$ for week 1. The resulting weekly-mean rate is clipped at zero (negative values, which can arise from numerical noise or an accumulation-counter reset in the archive, are set to zero rather than reported as negative precipitation); this clip is rarely binding in practice but is disclosed here as it is not visually distinguishable from an unclipped value in the result tables. The ERA5 and IMD truth references are constructed from true 24-hour daily totals and then averaged over the same lead window.

4.2 Anomalies and climatology

Both forecast and observed anomalies are defined by removing a single common climatology, applied identically to every system and to the truth: a 30-year (1990–2019) ERA5 day-of-year climatology for Z500 and ERA5-referenced precipitation, and the IMD 1991–2020 daily rainfall climatology (Pai et al., 2014) for IMD-referenced precipitation. Because each climatology is computed over decades independent of the verification season, it avoids the amplitude inflation that a same-season baseline would introduce.

The precipitation climatology always matches the precipitation truth source, never the reverse. JFM 2026 precipitation is verified against ERA5 (Section 2.2), so JFM precipitation anomalies use the ERA5 day-of-year climatology, converted from m to mm day^{-1}. JJAS 2019 precipitation is verified against IMD, so JJAS precipitation anomalies use the IMD daily climatology. This pairing is enforced in the verification pipeline for every model and every initialization; no run mixes an ERA5 truth with an IMD climatology or vice versa. This matters because anomaly correlation is sensitive to the climatology baseline: pairing the wrong climatology with a truth source would inflate or deflate ACC by introducing a spurious mean offset, independent of forecast quality. We flag this explicitly because it is the most common silent error in cross-reference S2S verification, and because the two seasons in this study deliberately use *different* truth sources (Section 2.2), so getting this pairing wrong in either direction would have been easy.

4.3 Deterministic and probabilistic metrics

The anomaly correlation coefficient (ACC; cf. Murphy, 1988) at lead ℓ is the cosine-latitude-weighted, spatially centered Pearson correlation of forecast and observed anomaly fields, computed per initialization and averaged over the season:

$$\mathrm{ACC}(\ell) = \frac{1}{T}\sum_{t=1}^{T} \frac{\sum_i w_i\,(f'_{i,t,\ell} - \overline{f'}_{t,\ell})\,(o'_{i,t,\ell} - \overline{o'}_{t,\ell})}{\sqrt{\sum_i w_i\,(f'_{i,t,\ell} - \overline{f'}_{t,\ell})^2}\,\sqrt{\sum_i w_i\,(o'_{i,t,\ell} - \overline{o'}_{t,\ell})^2}}\,, \tag{2}$$

where $w_i = \cos\varphi_i$ and $\overline{x}_{t,\ell}$ is the cosine-weighted spatial mean over land points. RMSE and bias are computed on the weekly-mean fields in their native units (mm day^{-1} for precipitation, m for Z500):

$$\mathrm{RMSE}(\ell) = \frac{1}{T}\sum_{t=1}^{T}\sqrt{\frac{\sum_i w_i\,(f_{i,t,\ell} - o_{i,t,\ell})^2}{\sum_i w_i}}\,, \qquad \mathrm{Bias}(\ell) = \frac{1}{T}\sum_{t=1}^{T}\frac{\sum_i w_i\,(f_{i,t,\ell} - o_{i,t,\ell})}{\sum_i w_i}\,. \tag{3}$$

For probabilistic verification, each system contributes an ensemble mean μ and spread (standard deviation) σ at each grid point and lead day. The continuous ranked probability score (CRPS; Gneiting and Raftery, 2007) is computed differently depending on the native form of each system's forecast, rather than forcing every system onto a single Gaussian summary. For the five systems delivered as individual members (FuXi-S2S, ECMWF, UKMO, NCEP, DLESyM), CRPS is the empirical finite-sample estimator (Hersbach, 2000),

$$\mathrm{CRPS}_{\mathrm{ens}} = \frac{1}{N}\sum_{n=1}^{N}|x_n - y| - \frac{1}{2N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}|x_n - x_m|\,, \tag{4}$$

computed directly from the N archived members x_n and the observation y, with no Gaussian assumption. Spire AI-S2S is delivered only as a mean and standard deviation, with no individual members available, so it alone is scored with the closed-form Gaussian CRPS,

$$\mathrm{CRPS}_{\mathrm{Gauss}}(\mu, \sigma, y) = \sigma\Big[\omega\,(2\Phi(\omega) - 1) + 2\phi(\omega) - \tfrac{1}{\sqrt{\pi}}\Big]\,, \quad \omega = \tfrac{y-\mu}{\sigma}, \tag{5}$$

with Φ, ϕ the standard-normal CDF and PDF. Because only Spire is scored this way, we checked directly how much this Gaussian convention—rather than a genuine skill difference—could move a score, by additionally scoring the member-based systems (ECMWF, FuXi-S2S, NCEP) with the Gaussian CRPS built from their own ensemble mean and spread and comparing it, cell for cell, to their empirical member CRPS on the JFM 2026 data. For Z500 the two agree closely (median absolute difference 0.9%, at most 2.7% across leads), so the Gaussian convention is immaterial for Z500 and all Z500 CRPSS comparisons involving Spire are robust to it. For precipitation, whose grid-point distribution is skewed and bounded below by zero, the Gaussian CRPS is systematically *larger* than the empirical CRPS (by ≈20% in the mean, growing with lead); the convention therefore mildly *penalizes* the system it is applied to, so Spire's precipitation CRPSS reported below is, if anything, a conservative estimate of its probabilistic precipitation skill relative to the member-based systems (Section 5.5, Section 7). Ensemble spread is clipped to a small per-variable floor ($\sigma \geq 0.05\,\mathrm{mm\,day}^{-1}$ for precipitation, 1 m for Z500) before use in Eq. (5) and in the spread–skill ratio below, to avoid ill-conditioned scores at grid points with near-zero ensemble spread; this floor is well below the smallest spreads encountered in practice and does not materially affect the reported scores. The skill score CRPSS $= 1 - \mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{clim}}$ is taken against a deterministic-climatology reference, $\mathrm{CRPS}_{\mathrm{clim}} = \overline{|c-y|}$, the cosine-weighted mean absolute error of the climatological field c against the observation—not a Gaussian climatological distribution—so CRPSS values reported here should be read as skill relative to a zero-spread climatology baseline rather than a calibrated probabilistic one; because every system shares the same reference, cross-system CRPSS *rankings* are unaffected; only the absolute CRPSS level is comparatively generous. Calibration is summarized by the spread–skill ratio SSR $= \overline{\sigma}/\mathrm{RMSE}(\mu)$, which equals one for a perfectly reliable ensemble and is below one when the ensemble is overconfident (under-dispersive). Brier scores for precipitation exceedance events use the empirical ensemble exceedance fraction for the five member-based systems and the Gaussian exceedance probability $P(y > \tau) = 1 - \Phi\big(\frac{\tau-\mu}{\sigma}\big)$ for Spire, for thresholds $\tau = 1$ and $10\,\mathrm{mm\,day}^{-1}$.

Multi-model ensemble (MME). The MME reported throughout this paper is the unweighted mean of the ensemble-mean fields of whichever individual systems successfully scored for a given variable, lead week, and initialization (a minimum of two contributing systems is required). Its membership is therefore not fixed: it excludes DLESyM for precipitation (no precipitation channel), excludes Spire for JJAS 2019 (unavailable that season), and can vary cell-to-cell wherever an individual system fails to open or score (Section 7). The JFM 2026 MME accordingly includes Spire alongside the other five systems, so MME comparisons in that season are not fully independent of Spire's own performance.

4.4 Initialization sets and matched comparison

The two seasons use different initialization protocols, reflecting both data availability and the goal of a fair, common-init comparison within each season.

- **JFM 2026:** 90 daily 00 UTC initializations (1 January–31 March 2026) for the six-system comparison. The ERA5 daily truth record used for JFM extends through 12 May 2026, covering the full week-6 (day-42) verification window of the latest (31 March) initialization, so all six lead weeks are scored on the complete, identical 90-initialization sample (Section 7). Using all 90 daily initializations, rather than a sparser weekly subset, gives a larger and more representative sample of the season; scores reported here are consequently more conservative (closer to the seasonal mean, less sensitive to any single favorable initialization) than an estimate from a small weekly subsample of the same season would be.
- **JJAS 2019:** 35 common Monday/Thursday initializations (3 June–30 September 2019) for the main monsoon comparison among ECMWF, UKMO, NCEP, and FuXi-S2S. Precipitation is verified against IMD rainfall and Z500 against ERA5. This 35-date set is the primary JJAS benchmark because it gives the largest exact common-initialization sample including FuXi-S2S and the three operational systems. DLESyM is not part of the main monsoon precipitation comparison because it lacks a precipitation channel; its Z500 behavior is described only as a smaller DLESyM-inclusive sensitivity comparison (Section 5.9).

4.5 Sampling uncertainty and significance

Every headline skill difference in this paper is a point estimate from a single season's finite sample of initializations, so we attach sampling uncertainty to it with a *paired* bootstrap over initialization dates (cf. Wilks, 2019). Because the season ACC is defined as the mean over initializations of the per-initialization spatial ACC (Eq. 2), and CRPSS is aggregated as the mean of the per-initialization skill score, resampling the initialization index and re-averaging reproduces exactly the statistic reported in the tables. For each of $B = 10{,}000$ resamples we draw the season's initialization dates with replacement and apply the *same* resampled index to every system, which preserves the strong between-system correlation across initializations (an easy winter for the atmosphere is an easy winter for all models at once) and thereby tests each pairwise skill difference far more powerfully than comparing two independently constructed error bars would. We report 95% percentile confidence intervals for each system's per-lead ACC and, for the model-to-model comparisons the paper actually makes, the mean paired difference with a two-sided bootstrap p-value and an indication of whether its 95% interval excludes zero. This is a deliberately conservative, distribution-free treatment: it quantifies robustness to *which initializations happened to fall in this one season*, but it cannot, of course, quantify the additional and larger uncertainty of having sampled only a single winter and a single monsoon (Section 7).

5 Results

5.1 Overview: skill across variables and seasons

Figure 2 summarizes the core benchmark: anomaly correlation versus lead week for precipitation and Z500 in the two case studies. The top row shows the JFM 2026 Spire-inclusive winter comparison; the bottom row shows the JJAS 2019 35-date monsoon comparison among ECMWF, UKMO, NCEP, FuXi-S2S, and the MME. Three features stand out immediately. First, in JFM 2026, Spire AI-S2S sits above every other individual system for precipitation at every lead, while its Z500 skill decays more gradually than FuXi-S2S and DLESyM. Second, in JJAS 2019, precipitation skill is much weaker and collapses rapidly: no individual model retains positive all-India precipitation ACC beyond week 3. Third, the physical contrast between seasons is larger than the model ranking alone: the winter case preserves a clearer circulation-to-rainfall signal, whereas the monsoon case loses rainfall skill even when some Z500 skill remains. The following subsections quantify each of these features in turn.

5.2 Winter (JFM 2026): the ML advantage

Precipitation. Spire AI-S2S has the highest precipitation ACC of all six systems at every lead week (Table 2): 0.78 at week 1, versus 0.73 for both ECMWF and UKMO and 0.62 for NCEP, narrowing to 0.31 by week 6 where it remains tied with the multi-model ensemble (MME) for the top score. The paired bootstrap (Section 4.5, Table 5) shows this lead is not uniform in strength across the lead range. Spire's advantage over ECMWF is statistically significant at *every* lead week (mean paired ACC difference +0.03 to +0.08; 95% interval excludes zero at all six weeks), and its lead over every precipitation-capable individual system is significant through week 3. From week 4 onward Spire remains the highest-scoring individual system by point estimate, but its margin over FuXi-S2S and UKMO falls within the paired-bootstrap confidence interval (differences of 0.00–0.04), while its lead over NCEP remains significant at weeks 4 and 6 (+0.05 each) though not at week 5. At weeks 4–6 Spire is therefore clearly best against ECMWF, intermittently so against NCEP, and not distinguishable from FuXi-S2S or UKMO; the MME remains nearly identical by point estimate. Spire also has the lowest RMSE at every lead (Table 3; 1.09 mm day^{-1} at week 1, rising to 2.28 mm day^{-1} at week 6, against 1.27–1.57 and 2.47–2.72 mm day^{-1} respectively for the other individual systems), so the ACC advantage is not an artifact of a smoother forecast field receiving a more favorable correlation score. FuXi-S2S, after the unit harmonization of Section 3.2, is competitive with the operational dynamical systems (ACC 0.72 at week 1) but does not match Spire at any lead.

Table 2: JFM 2026 all-India precipitation anomaly correlation (ACC) by lead week, verified against ERA5 daily totals. Best individual-model score in each column is bold; MME is the multi-model mean.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	**0.78**	**0.66**	**0.54**	**0.42**	**0.36**	**0.31**
FuXi-S2S	0.72	0.55	0.45	0.39	0.35	0.31
ECMWF	0.73	0.61	0.46	0.37	0.32	0.27
UKMO	0.73	0.50	0.39	0.38	0.35	0.28
NCEP	0.62	0.49	0.40	0.36	0.34	0.26
MME	0.74	0.61	0.49	0.42	0.37	0.31

Table 3: JFM 2026 all-India precipitation RMSE (mm day^{-1}) by lead week.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	**1.09**	**1.44**	**1.68**	**1.88**	**2.14**	**2.28**
FuXi-S2S	1.33	1.83	1.95	2.08	2.34	2.47
ECMWF	1.27	1.59	1.92	2.14	2.40	2.57
UKMO	1.31	1.90	2.16	2.14	2.36	2.56
NCEP	1.57	1.93	2.15	2.29	2.49	2.75
MME	1.18	1.54	1.80	1.94	2.18	2.35

Z500. The clearest separation in the entire benchmark appears in Z500 (Table 4). At week 1 the field is tightly clustered (ACC 0.89–0.97 across all six systems, with ECMWF and the MME marginally ahead at 0.97), and all three ML systems are essentially indistinguishable from the dynamical systems. By week 3 the field has split sharply: Spire, ECMWF, UKMO, and NCEP all remain skillful (ACC 0.40–0.66), while FuXi-S2S has fallen to 0.11 and DLESyM to 0.25. By week 4 both FuXi-S2S and DLESyM have crossed into *negative* ACC (−0.15 and −0.19) and remain negative through week 6 (−0.18 and −0.33)—meaning their week-4–6 Z500 forecasts are anticorrelated with the true anomaly pattern more often than not. Every system, including Spire, loses further skill between week 4 and week 6: none of the six holds Z500 skill flat to the end of the forecast horizon. The distinguishing feature of Spire (and, less consistently, NCEP) is not the absence of decay but its slower rate: Spire is the highest- or second-highest individual system at every lead from week 3 onward (ACC 0.39–0.51 at weeks 4–6, versus ECMWF's fall to 0.13 by week 6 and FuXi-S2S/DLESyM's collapse into negative territory), so it degrades more gracefully than either the operational systems or the other two ML systems, without being immune to lead-time decay itself. The paired bootstrap confirms that Spire's Z500 advantage over the two other ML systems is significant at every lead and large after week 1: Spire exceeds FuXi-S2S and DLESyM at every lead with 95% intervals that exclude zero (Table 6), and the negative week-4–6 FuXi-S2S and DLESyM ACC values are themselves significantly below zero. Spire and ECMWF, by contrast, separate significantly only at the two ends of the horizon—in ECMWF's favour at week 1 and in Spire's at week 6—and are statistically indistinguishable in Z500 from week 2 through week 5, consistent with describing Spire as the most *consistently* skillful system rather than the uniquely best one at every individual lead.

Table 4: JFM 2026 all-India Z500 ACC by lead week (ERA5 truth).

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.94	0.85	**0.66**	**0.51**	0.39	**0.41**
FuXi-S2S	0.89	0.56	0.11	−0.15	−0.27	−0.18
DLESyM	0.92	0.62	0.25	−0.19	−0.36	−0.33
ECMWF	**0.97**	**0.86**	0.65	0.49	0.32	0.13
UKMO	0.96	0.74	0.51	0.41	0.39	0.32
NCEP	0.90	0.63	0.40	0.41	**0.41**	0.37
MME	0.97	0.86	0.63	0.41	0.40	0.32

Table 5: JFM 2026 precipitation: paired-bootstrap significance of Spire AI-S2S's all-India ACC lead over each other system, by lead week.

vs. Spire AI-S2S	W1	W2	W3	W4	W5	W6
FuXi-S2S	$+0.06^*$	$+0.11^*$	$+0.09^*$	$+0.02^\dagger$	$+0.01^\dagger$	$0.00^\dagger$
ECMWF	$+0.04^*$	$+0.04^*$	$+0.08^*$	$+0.05^*$	$+0.03^*$	$+0.04^*$
UKMO	$+0.04^*$	$+0.15^*$	$+0.14^*$	$+0.04^\dagger$	$0.00^\dagger$	$+0.03^\dagger$
NCEP	$+0.16^*$	$+0.17^*$	$+0.14^*$	$+0.05^*$	$+0.02^\dagger$	$+0.05^*$

Mean paired difference in ACC (Spire AI-S2S minus competitor) over the season's initializations; positive favours Spire AI-S2S. *: 95% paired-bootstrap confidence interval (10,000 resamples of initialization dates) excludes zero; †: interval includes zero (difference not distinguishable from sampling noise).

5.3 Monsoon (JJAS 2019): the great equalizer

Precipitation. The monsoon precipitation benchmark is qualitatively different from the JFM winter case (Table 7). On the 35 common Monday/Thursday initializations, week-1 all-India ACC is only about 0.39 for ECMWF, UKMO, and the MME, 0.28 for NCEP, and 0.25 for FuXi-S2S. By week 3, every individual system has fallen to ACC ≤ 0.04, and by week 4 every individual system is non-positive. The paired bootstrap sharpens this into a cleaner statement: only weeks 1 and 2 have all-India precipitation ACC whose 95% interval excludes zero for the leading systems, and by week 3 no individual system's ACC is significantly different from zero (Section 4.5). Thus the central monsoon result is not that one model class beats another, but that weekly rainfall anomalies over India lose *statistically detectable* deterministic skill by week 3 for all available systems in this case. This is a qualitatively different outcome from JFM 2026, where several systems retained positive precipitation skill through week 6.

Z500. Z500 retains skill longer than precipitation, but the ranking differs from the JFM case (Table 8). ECMWF is the most skillful individual system at every lead in the 35-date comparison: ACC is 0.91 at week 1, 0.37 at week 3, and remains positive at 0.25–0.29 by weeks 5–6. UKMO is the second strongest early-lead dynamical system, while FuXi-S2S is substantially weaker on the 35-date set (0.68 at week 1 and 0.13 at week 3). The contrast with precipitation implies that the monsoon failure is not simply a complete loss of large-scale circulation information; rather, useful weekly rainfall skill is lost faster than Z500 skill, consistent with the difficulty of translating monsoon circulation anomalies into regional rainfall.

Table 6: JFM 2026 Z500: paired-bootstrap significance of Spire AI-S2S's all-India ACC difference against each other system, by lead week.

vs. Spire AI-S2S	W1	W2	W3	W4	W5	W6
FuXi-S2S	$+0.05^{*}$	$+0.29^{*}$	$+0.55^{*}$	$+0.65^{*}$	$+0.66^{*}$	$+0.59^{*}$
DLESyM	$+0.02^{*}$	$+0.23^{*}$	$+0.41^{*}$	$+0.70^{*}$	$+0.75^{*}$	$+0.74^{*}$
ECMWF	-0.03^{*}	$-0.01^{\dagger}$	$+0.01^{\dagger}$	$+0.02^{\dagger}$	$+0.08^{\dagger}$	$+0.28^{*}$
UKMO	-0.02^{*}	$+0.11^{*}$	$+0.15^{*}$	$+0.10^{\dagger}$	$0.00^{\dagger}$	$+0.09^{\dagger}$
NCEP	$+0.04^{*}$	$+0.22^{*}$	$+0.26^{*}$	$+0.10^{*}$	$-0.01^{\dagger}$	$+0.04^{\dagger}$

Mean paired difference in ACC (Spire AI-S2S minus competitor) over the season's initializations; positive favours Spire AI-S2S. *: 95% paired-bootstrap confidence interval (10,000 resamples of initialization dates) excludes zero; †: interval includes zero (difference not distinguishable from sampling noise).

Table 7: JJAS 2019 all-India precipitation ACC by lead week, verified against IMD gridded rainfall (35 common Monday/Thursday initializations).

Model	W1	W2	W3	W4	W5	W6
FuXi-S2S	0.25	0.09	−0.03	−0.05	−0.07	−0.08
ECMWF	**0.39**	0.17	0.03	**0.00**	**−0.02**	**−0.04**
UKMO	0.39	**0.20**	**0.04**	−0.07	−0.07	−0.09
NCEP	0.28	0.07	−0.02	−0.11	−0.11	−0.09
MME	0.39	0.17	0.01	−0.07	−0.09	−0.09

5.4 Divergence among the ML systems

The JFM 2026 Z500 results contain a finding that is easy to miss when comparing ML systems only to dynamical baselines: the ML systems do not behave like a single class of model. At week 1, Spire, FuXi-S2S, and DLESyM are all skillful and mutually close (Z500 ACC 0.94, 0.89, and 0.92 respectively; Table 4)—a week-1 leaderboard would rank the three ML systems as roughly interchangeable. By week 4 this is no longer true: Spire is at 0.51 and remains positive through week 6 (0.41), while FuXi-S2S and DLESyM have already crossed into negative territory at week 4 (−0.15 and −0.19) and stay negative through week 6.

The practical implication is that *long-lead stability is a separate property from headline (week-1) skill.* Spire still loses skill with lead (Section 5.2), but it does not collapse the way FuXi-S2S and DLESyM do in the JFM case. FuXi-S2S and DLESyM, despite being built on different architectures (transformer and diffusion-based, respectively) and trained on different data, share a qualitatively similar long-lead failure mode in this winter benchmark—turning anticorrelated with the truth rather than merely losing skill. We do not have access to the internals of either system and so cannot attribute this directly to a specific architectural or training choice, but the pattern is consistent with the broader concern that ML systems optimized for short-range accuracy can develop compounding error growth once integrated well beyond their primary training horizon (Section 6). The practical caution for anyone selecting an S2S system based on a short evaluation window is that a single week-1 score is not sufficient evidence of usable skill at weeks 4–6.

This divergence is also spatially coherent, not an artifact of the domain average. Figure 3 maps the per-grid-point temporal anomaly correlation (the correlation across the 90 initializations between each system's Z500 anomaly and the ERA5 anomaly at that cell) at week 3. Spire and ECMWF retain positive local correlation across essentially the whole of India; FuXi-S2S

Table 8: JJAS 2019 all-India Z500 ACC by lead week (ERA5 truth, 35 common Monday/Thursday initializations).

Model	W1	W2	W3	W4	W5	W6
FuXi-S2S	0.68	0.35	0.13	0.18	0.02	0.07
ECMWF	**0.91**	**0.64**	**0.37**	**0.36**	**0.25**	**0.29**
UKMO	0.88	0.53	0.21	0.16	0.12	0.08
NCEP	0.71	0.26	0.08	0.00	0.04	0.11
MME	0.89	0.56	0.23	0.22	0.17	0.16

and, more so, DLESyM are already weak and patchy by week 3; and NCEP develops outright *negative* local correlation over central and peninsular India. The ML systems thus diverge coherently in space, reinforcing that the long-lead separation among them is a real skill difference rather than a cancellation of regional signals in the domain mean.

5.5 Probabilistic skill and calibration

The deterministic ranking of Section 5.2 does not simply carry over to probabilistic skill. For precipitation (Table 9; Figure 4a), Spire has the highest CRPSS at week 1 (0.61, narrowly ahead of ECMWF's 0.59), but ECMWF overtakes it from week 2 through week 5 (e.g. 0.51 versus 0.40 at week 2, 0.30 versus 0.18 at week 5), while FuXi-S2S narrowly has the highest week-6 CRPSS. This precipitation ranking should, however, be read with the scoring-convention check of Section 4.3 in mind: because Spire is scored with the Gaussian CRPS, which inflates the score by ≈20% for skewed precipitation relative to the empirical member CRPS used for ECMWF, part of ECMWF's apparent week-2–5 precipitation-CRPSS lead is attributable to the convention rather than to a genuine probabilistic-skill gap; the Z500 CRPSS comparison below, where the convention is immaterial (< 3%), is the more convention-robust probabilistic result. For Z500 (Table 10; Figure 4b), ECMWF leads more clearly at short lead (CRPSS 0.82 at week 1 versus 0.59 for Spire) and remains ahead through week 5, but the gap narrows steadily with lead (0.38 versus 0.29 at week 3, 0.26 versus 0.26 tied at week 4, 0.24 versus 0.23 at week 5) and Spire edges ahead at week 6 (0.24 versus 0.22), consistent with its more graceful deterministic Z500 decay at the same leads (Section 5.4). FuXi-S2S and NCEP show the least stable CRPSS trajectories: FuXi-S2S Z500 CRPSS falls to −0.23 at week 3 before partially recovering, and NCEP Z500 CRPSS is negative from week 2 onward and reaches −0.30 by week 6, the worst probabilistic Z500 trajectory of any system.

Figure 5 shows why ECMWF's probabilistic precipitation skill exceeds its deterministic ranking. Every system is under-dispersive (SSR < 1, indicating overconfidence) at week 1—Spire is closest to calibrated (SSR 0.63) while FuXi-S2S is the most overconfident (SSR 0.11). By week 3, ECMWF's ensemble has grown to near-ideal calibration for precipitation (SSR 0.70) while Spire has *over*-corrected into over-dispersion (SSR 1.43, indicating its spread now exceeds its error). For Z500 the same pattern is sharper still: Spire's spread–skill ratio rises to 1.75 by week 4 and stays above 1.3 through week 6, meaning its ensemble spread is substantially larger than its actual error from mid-range lead onward—a sign of growing overdispersion that does not by itself harm CRPSS (an overdispersive ensemble can still be skillful) but indicates that Spire's stated uncertainty should not be taken as quantitatively calibrated at long lead.

Table 9: JFM 2026 all-India precipitation CRPSS versus climatology by lead week. Positive values indicate probabilistic skill above climatology.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	**0.61**	0.40	0.25	0.19	0.18	0.17
FuXi-S2S	0.45	0.27	0.31	0.32	0.29	**0.26**
ECMWF	0.59	**0.51**	**0.40**	**0.35**	**0.30**	0.25
UKMO	0.45	0.27	0.18	0.25	0.23	0.18
NCEP	0.42	0.33	0.29	0.26	0.24	0.14

Table 10: JFM 2026 all-India Z500 CRPSS versus climatology by lead week.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.59	0.42	0.29	0.26	0.23	**0.24**
FuXi-S2S	0.68	0.10	−0.23	−0.20	0.03	0.13
DLESyM	0.54	0.30	0.10	0.01	0.11	0.11
ECMWF	**0.82**	**0.60**	**0.38**	**0.26**	**0.24**	0.22
UKMO	0.76	0.34	0.12	0.14	0.08	0.07
NCEP	0.27	−0.14	−0.22	−0.27	−0.29	−0.30

ECMWF's Z500 spread–skill ratio is the closest to one of any system across most of the lead range (0.89 at week 1, peaking at 1.39 by week 3, settling to 1.04 by week 6), consistent with its large ensemble and operational calibration heritage, though it too overshoots unity at mid-range lead rather than holding exactly at one throughout.

5.6 Bias analysis

ACC and CRPSS measure pattern agreement and probabilistic skill but not the sign or magnitude of systematic error, which matters operationally (e.g. a forecast that is reliably too wet or too dry is correctable even if its pattern skill is otherwise good). Table 11 shows that Spire's precipitation bias is the smallest in magnitude at most leads and changes sign across the lead range (−0.04 at week 1, peaking at +0.12 at week 3, back to −0.01 by week 6), rather than drifting monotonically in one direction. ECMWF and FuXi-S2S both develop a growing *dry* bias with lead ($-0.15 \to -0.44$ and $-0.14 \to -0.51\,\mathrm{mm\,day}^{-1}$ respectively, week 1 to week 6), consistent with the general tendency of MSE-trained forecast systems to smooth toward the climatological mean at long lead. UKMO carries a small, fairly stable *wet* bias throughout (+0.04 to +0.22).

For Z500 (Table 12), the systems separate into two clear groups. Spire and DLESyM start with a sizeable *positive* height bias at week 1 (+11.3 and +8.8 m) that decays with lead; DLESyM crosses zero by week 3 and is the system closest to zero bias at weeks 3–5 (−0.9 to −4.9 m), while Spire's bias decays more slowly and remains positive throughout (+6.0 to +7.6 m at weeks 4–6). ECMWF, UKMO, and especially NCEP and FuXi-S2S instead develop a *negative* height bias with lead (NCEP reaches −29 m by week 4 and stays there; FuXi-S2S reaches −23 m at weeks 3–4 before partially recovering to −9 m by week 6). UKMO is closest to zero at week 1 (−0.3 m) and again at week 6 (+1.3 m). No single system is closest to zero across the entire lead range: DLESyM's small mid-range bias does not carry over to its ACC, which collapses at the same leads (Section 5.2)—a reminder that low bias and high pattern correlation are distinct

Table 11: JFM 2026 all-India precipitation bias (mm day^{-1}, forecast minus ERA5) by lead week. Bold = closest to zero.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	**−0.04**	+0.10	**+0.12**	+0.11	+0.03	**−0.01**
FuXi-S2S	−0.14	**−0.02**	−0.15	−0.29	−0.38	−0.51
ECMWF	−0.15	−0.11	−0.16	−0.18	−0.34	−0.44
UKMO	+0.05	+0.19	+0.22	+0.15	+0.06	+0.04
NCEP	−0.19	−0.09	−0.17	**−0.07**	**−0.01**	+0.07
MME	−0.10	+0.02	−0.03	−0.06	−0.13	−0.17

Table 12: JFM 2026 all-India Z500 bias (m, forecast minus ERA5) by lead week. Bold = closest to zero.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	+11.34	+9.38	+7.07	+5.99	+6.88	+7.62
FuXi-S2S	−2.15	−14.10	−23.06	−23.31	−16.40	−8.73
DLESyM	+8.77	+4.25	**−0.88**	**−4.14**	**−4.94**	−1.78
ECMWF	+3.81	**−1.54**	−6.47	−9.15	−6.19	−2.23
UKMO	**−0.32**	−4.27	−8.92	−7.49	−4.94	**+1.26**
NCEP	−14.88	−25.65	−28.72	−29.49	−27.92	−26.95
MME	+1.09	−5.32	−10.16	−11.26	−8.92	−5.14

properties and need not coincide.

5.7 Regional structure of skill

Figure 9 and Table 13 break down week-1 JFM 2026 precipitation ACC by the four IMD homogeneous regions (Section 2.1). Predictability is markedly unequal across India: Central India, East/Northeast India, and Northwest India are all well-predicted at week 1 by every system (ACC 0.61–0.80), while the South Peninsula is uniformly the hardest region (ACC 0.20–0.46 across all six systems)—roughly half the skill level of the other three regions for every model without exception. This regional gap is consistent with the synoptic character of winter precipitation in northern/central India versus the more locally forced, lower-amplitude rainfall the peninsula receives outside the northeast-monsoon season. Spire's domain-mean lead over the other systems (Section 5.2) is not concentrated in any one region: at week 1 it is tied or within 0.01 of ECMWF in Central India (0.75 vs. 0.79) and Northwest India (0.80 vs. 0.79), and leads outright in East/Northeast India (0.80 vs. 0.75) and the South Peninsula (0.46, tied with ECMWF). No system—including Spire—shows a clear advantage specific to the hardest (South Peninsula) region, indicating that the winter ML advantage documented above is a broad-domain property rather than a regional artifact. The same regional split is visible for Z500 at week 1: every system scores ACC 0.69–0.96 across all four regions, and even the hardest region (South Peninsula) remains highly skillful for every system (ACC 0.69–0.88), a much narrower spread than for precipitation, reflecting the larger spatial decorrelation scale of the 500-hPa height field relative to rainfall.

The full lead-time evolution of regional precipitation skill (Figure 6) refines this week-1 snapshot. Spire's regional lead is clearest and most persistent in East/Northeast India, where it stays above the other systems through week 3; in Central India the operational systems

Table 13: JFM 2026 week-1 precipitation ACC by IMD homogeneous region. Best individual-model score in each column is bold.

Model	Northwest	Central	South Pen.	East/NE
Spire AI-S2S	**0.80**	0.75	**0.46**	**0.80**
FuXi-S2S	0.69	0.74	0.38	0.75
ECMWF	0.79	**0.79**	0.46	0.75
UKMO	0.75	0.71	0.38	0.75
NCEP	0.61	0.69	0.20	0.70
MME	0.78	0.77	0.37	0.76

(notably NCEP and ECMWF) catch up to and slightly overtake it at weeks 4–6, and in the South Peninsula all systems decay to a common, low skill level within two to three weeks. The regional panels also make explicit that no system holds useful (ACC $\gtrsim 0.5$) precipitation skill past week 2–3 in any single region, so the domain-mean decay of Section 5.2 is not hiding a region that stays strongly predictable at long lead.

Averaging over the full week 1–6 lead range gives the season-long regional picture (Table 14). For precipitation, Spire leads or ties in three of four regions (Northwest 0.43, East/NE 0.55, narrowly tying ECMWF in the South Peninsula at 0.38) but is overtaken by NCEP and ECMWF in Central India (0.56 versus 0.57 each); the All-India mean nonetheless favors Spire (0.51) because the other systems' Central India strength does not offset their weaker performance elsewhere. For Z500, Spire leads in two of four regions (Northwest 0.54, Central 0.55) and is competitive in East/NE (0.53 versus ECMWF's region-leading 0.59), but ECMWF leads outright in the South Peninsula (0.45) and East/NE, so the seasonal-mean Z500 picture is more evenly split between Spire and ECMWF than the week-1–3 results alone would suggest—consistent with the more graceful-but-not-flat long-lead decay documented in Section 5.4. RMSE (Table 15) shows a similar split: Spire has the lowest All-India precipitation RMSE ($1.75\,\mathrm{mm\,day^{-1}}$) and is best in two of four regions, but ECMWF has the lowest All-India and regional Z500 RMSE everywhere except Northwest India, where Spire is marginally better. Bias (Table 16) again shows East/Northeast India as the region where every system except UKMO under-forecasts precipitation substantially (Spire −0.88; FuXi-S2S and ECMWF −1.64; NCEP −1.33), a known orographic-smoothing effect at 1.5° resolution; UKMO is the exception, with a small positive bias there (+0.30) rather than the dry bias every other system shares. For Z500, no single system is closest to zero bias in every region: DLESyM is closest at the All-India and Central scales (+0.2 and +4.8 m), Spire is closest in Northwest India (+5.6 m), UKMO in the South Peninsula (−0.3 m), and ECMWF in East/Northeast India (+1.8 m)—bias and pattern skill are clearly separate properties at the regional scale, just as they are domain-wide (Section 5.6).

The monsoon precipitation collapse of Section 5.3 is also a broad-domain property rather than a regional artifact (Table 17). At week 1, JJAS 2019 regional skill is already lower and flatter than JFM 2026: ACC ranges only 0.07–0.45 across all four regions and the four individual systems in the 35-date JJAS benchmark, versus the much wider 0.20–0.80 range in JFM (Table 13). By week 3 every region for every system has fallen to ACC ≤ 0.16, with most region–model combinations at or below zero; no region retains the kind of useful skill that Northwest, Central, and East/Northeast India still show in JFM 2026 at the same lead.

Table 14: JFM 2026 region-wise ACC by IMD homogeneous region, weeks 1–6 mean (90-initialization average). Best system per region in bold.

Region		Spire AI-S2S	FuXi-S2S	DLESyM	ECMWF	UKMO	NCEP	MME
Precipitation	All India	**0.51**	0.46	–	0.46	0.44	0.41	0.49
	Northwest	**0.43**	0.42	–	0.39	0.39	0.29	0.42
	Central	0.56	0.50	–	0.57	0.45	**0.57**	0.57
	S. Peninsula	0.38	0.37	–	**0.38**	0.35	0.32	0.37
	East/NE	**0.55**	0.50	–	0.52	0.44	0.49	0.52
Z500	All India	**0.63**	0.16	0.15	0.57	0.56	0.52	0.60
	Northwest	**0.54**	0.21	0.23	0.49	0.47	0.48	0.54
	Central	**0.55**	0.17	0.13	0.53	0.50	0.46	0.51
	S. Peninsula	0.39	0.36	0.22	**0.45**	0.39	0.25	0.41
	East/NE	0.53	0.19	0.23	**0.59**	0.47	0.43	0.57

Table 15: JFM 2026 region-wise RMSE by IMD homogeneous region, weeks 1–6 mean. Precipitation in mm day^{-1}; Z500 in m. Best (lowest) system per region in bold.

Region		Spire AI-S2S	FuXi-S2S	DLESyM	ECMWF	UKMO	NCEP	MME
Precipitation	All India	**1.75**	2.00	–	1.98	2.07	2.20	1.83
	Northwest	1.44	1.43	–	1.41	1.49	1.70	**1.37**
	Central	0.53	0.51	–	**0.43**	0.57	0.47	0.45
	S. Peninsula	**0.95**	1.04	–	0.97	1.14	1.06	0.97
	East/NE	**3.37**	4.06	–	4.07	4.26	4.36	3.70
Z500	All India	22.76	28.55	26.68	**21.67**	23.95	32.63	21.85
	Northwest	**26.93**	38.85	32.33	27.83	30.26	36.30	28.16
	Central	19.92	20.72	21.94	**17.77**	19.86	30.63	17.91
	S. Peninsula	14.57	11.72	17.68	**10.17**	11.29	22.77	10.25
	East/NE	20.62	24.99	21.56	**19.07**	22.81	31.23	19.36

East/Northeast India is a partial, model-dependent exception—ECMWF (0.16) and UKMO (0.14) retain some skill there at week 3 while collapsing elsewhere—but no system or region combination approaches the JFM 2026 winter skill level. The regional data therefore reinforce the domain-mean finding of Section 5.3: the monsoon equalizes skill across the whole of India, not just in the domain average.

5.8 Rainfall case studies: winter and monsoon

The aggregate scores above are complemented by a single concrete event, chosen as the strongest positive domain-mean precipitation anomaly of the JFM 2026 season: the wet spell verifying 15–21 March 2026, when a western disturbance produced heavy precipitation over northwest India and the Himalayan foothills together with rainfall over northeast India (Figure 7, top-left truth panel). Figure 7 shows the ERA5 truth alongside each system's forecast of this week at two leads: week-1 (initialized 14 March) and week-2 (initialized 7 March). At week-1 lead every system places the northwest-India rainfall maximum in broadly the correct location, with the dynamical systems (ECMWF, UKMO, NCEP) producing the sharpest local amplitudes and Spire a smoother, lower-amplitude field—consistent with its small negative precipitation bias and low RMSE (Sections 5.2, 5.6) rather than a displaced pattern. At week-2 lead all systems weaken and smooth the event, but the northwest-India signal survives in every system, illustrating at the level of a single event the same conclusion the seasonal ACC gives: winter

Table 16: JFM 2026 region-wise mean bias (forecast minus ERA5) by IMD homogeneous region, weeks 1–6 mean. Precipitation bias in mm day^{-1}; Z500 bias in m. Value closest to zero per region in bold.

Region		Spire AI-S2S	FuXi-S2S	DLESyM	ECMWF	UKMO	NCEP	MME
	All India	**+0.05**	−0.25	–	−0.23	+0.12	−0.08	−0.08
	Northwest	+0.23	**+0.02**	–	+0.04	−0.02	+0.38	+0.13
Precipitation	Central	+0.25	+0.05	–	−0.03	+0.17	**0.00**	+0.09
	S. Peninsula	+0.13	−0.14	–	+0.05	+0.12	**−0.01**	+0.03
	East/NE	−0.88	−1.64	–	−1.64	**+0.30**	−1.33	−1.04
	All India	+8.05	−14.62	**+0.21**	−3.63	−4.11	−25.60	−6.62
	Northwest	**+5.61**	−29.72	−10.64	−8.95	−8.13	−25.28	−12.85
Z500	Central	+7.69	−7.99	+4.78	−3.27	**−3.12**	−28.10	−5.00
	S. Peninsula	+10.96	+2.31	+14.91	+0.69	**−0.28**	−21.54	+1.17
	East/NE	+10.56	−17.90	−4.80	**+1.83**	−2.38	−25.65	−6.39

rainfall tied to organized western-disturbance circulation remains partially forecastable into the second week. The case study is a qualitative illustration and is not used to rank the systems; the season-long scores above remain the basis for every quantitative claim.

The monsoon counterpart (Figure 8) is the strongest active-monsoon spell of JJAS 2019, verifying 6–12 August 2019, when heavy rain fell along the Western Ghats and across central India (IMD truth, top-left panel). The event-level behaviour mirrors the seasonal scores of Section 5.3: at week-1 lead the dynamical systems (ECMWF, UKMO, NCEP) reproduce the Western-Ghats and central-India rainfall bands with roughly correct location if smoothed amplitude, whereas FuXi-S2S substantially under-represents the spell almost everywhere; by week-2 lead all systems weaken further and the correspondence with the observed pattern is largely lost. The contrast between the two case studies is the same one the aggregate ACC draws: the winter western-disturbance event stays partially forecastable into week 2, while the monsoon active spell is already poorly captured at week-1 lead by the ML system and only moderately by the dynamical systems—a single-event view of why monsoon rainfall is the harder subseasonal target. As with the winter case, these panels are qualitative illustrations, not an additional ranking.

5.9 DLESyM-inclusive monsoon sensitivity

The main JJAS 2019 comparison excludes DLESyM because DLESyM has no precipitation channel and because its monsoon availability is more limited than the operational-plus-FuXi set. As a sensitivity check, we also inspect the smaller 17-Thursday subset on which DLESyM, FuXi-S2S, ECMWF, UKMO, and NCEP are all available for Z500. This subset does not change the interpretation of the monsoon case. Precipitation skill for the systems with precipitation remains short-lived, falling to ACC ≤ 0.06 by week 3. DLESyM Z500 starts competitively at week 1 (ACC 0.85) but falls to near zero by week 3 (0.02) and negative by week 4 (−0.12). Thus adding DLESyM to a smaller matched monsoon subset reinforces, rather than weakens, the conclusion that the monsoon case does not preserve the JFM-style long-lead ML separation.

Table 17: JJAS 2019 precipitation ACC by IMD homogeneous region for every lead week 1–6 (IMD truth, 35 common Monday/Thursday initializations), showing that the monsoon skill collapse is uniform across regions: no region–model combination retains useful skill beyond week 2–3.

Region	Model	W1	W2	W3	W4	W5	W6
Northwest	FuXi-S2S	0.28	0.16	0.06	**0.10**	**0.03**	**0.00**
	ECMWF	**0.45**	**0.23**	**0.11**	0.09	0.03	−0.02
	UKMO	0.41	0.16	0.05	−0.04	0.03	−0.04
	NCEP	0.28	0.10	0.00	−0.01	−0.08	−0.02
	MME	0.42	0.19	0.06	0.02	−0.03	−0.03
Central	FuXi-S2S	0.19	0.02	−0.06	−0.12	−0.13	−0.11
	ECMWF	**0.33**	0.12	**0.03**	**−0.05**	**−0.06**	**−0.07**
	UKMO	0.32	**0.13**	−0.03	−0.11	−0.09	−0.09
	NCEP	0.26	0.04	−0.06	−0.13	−0.17	−0.17
	MME	0.32	0.09	−0.03	−0.12	−0.13	−0.12
South Pen.	FuXi-S2S	0.24	0.20	0.10	**0.06**	0.01	**0.02**
	ECMWF	0.31	0.21	0.08	0.00	−0.01	−0.01
	UKMO	**0.32**	**0.30**	**0.14**	0.01	−0.02	0.01
	NCEP	0.26	0.19	0.13	0.04	**0.01**	−0.03
	MME	0.31	0.24	0.12	0.03	−0.01	0.01
East/NE	FuXi-S2S	0.21	0.09	−0.01	−0.01	0.05	0.06
	ECMWF	**0.42**	**0.25**	**0.16**	0.11	0.08	**0.08**
	UKMO	0.30	0.12	0.14	**0.12**	**0.13**	0.08
	NCEP	0.07	−0.04	−0.08	−0.09	−0.05	0.01
	MME	0.35	0.15	0.09	0.05	0.06	0.08

6 Discussion

Where ML helps, and where it does not. The two-case design of this study lets us separate model ranking from physical predictability regime. In JFM 2026, winter precipitation over India is organized substantially by western disturbances—synoptic-scale systems with several days of coherent, predictable structure (Dimri et al., 2015; Hunt et al., 2018)—and Spire AI-S2S converts this structure into a clear precipitation advantage and comparatively slow Z500 decay (Section 5.2). In JJAS 2019, monsoon precipitation is dominated by smaller-scale convection embedded in a noisier large-scale envelope, and the 35-date benchmark shows that no available system retains useful weekly rainfall skill past week 2–3 (Section 5.3). The honest reading is therefore not that ML systems are generically skillful or unskillful, but that the physical regime controls what kind of skill can be extracted. The winter case demonstrates that AI S2S guidance can add value over India when large-scale circulation and rainfall are tightly coupled; the monsoon case shows that this value does not automatically generalize to the country's most consequential rainy season. These results should therefore be read as a case-season benchmark, not as a climatological skill estimate.

Not all AI systems are equal. A second contribution of this study is methodological: the JFM 2026 case shows that "ML skill" is not a single property. Spire, FuXi-S2S, and DLESyM have similar week-1 Z500 skill, but only Spire remains positively correlated with the truth through week 6 (Section 5.4). FuXi-S2S and DLESyM, despite different architectures (transformer and diffusion-based respectively) and different training regimes, share a similar long-lead failure mode that is not visible from their week-1 scores alone. We cannot identify the specific architectural or training cause of this divergence from the forecast output alone, but it is consistent with compounding error growth once an autoregressive or iteratively sampled model is pushed well beyond the lead range it was primarily optimized for. The practical consequence for anyone selecting or comparing ML S2S systems is that a single short-lead evaluation window is not sufficient evidence of skill at the leads that matter for subseasonal decision-making.

Deterministic versus probabilistic skill. The third finding is that ranking systems by ensemble-mean ACC and ranking them by CRPSS can disagree. ECMWF leads JFM precipitation CRPSS from weeks 2 to 5 despite Spire's higher ACC at every lead (Section 5.5), because ECMWF's 100-member ensemble, built on a mature operational reforecast-based calibration pipeline, achieves a spread–skill ratio closer to one across the full lead range, while Spire's delivered uncertainty becomes increasingly overdispersive beyond week 2. This deterministic-versus-probabilistic contrast holds most cleanly for Z500, where the empirical and Gaussian CRPS estimators agree to within a few percent; for precipitation, part of ECMWF's CRPSS lead reflects the Gaussian scoring convention applied to Spire rather than a pure skill gap (Sections 4.3, 7), so we lean on the Z500 result for the qualitative claim. The broader point survives either way: a forecast system that is the best deterministic choice is not automatically the best choice for probabilistic, risk-based decision-making (e.g. threshold-exceedance warnings), and operational guidance for India should evaluate candidate S2S systems on both axes rather than either alone.

Consistency with prior S2S literature. Multi-model S2S assessments over South Asia report useful deterministic precipitation skill confined to roughly weeks 1–2, with regime- and teleconnection-dependent extensions at longer leads (de Andrade et al., 2019; Pegion et al., 2019; Mariotti et al., 2020). Our results are consistent with this picture in both case studies: JJAS 2019 precipitation skill is effectively exhausted by week 2–3 for every available precipitation system in the 35-date monsoon benchmark, and even the most skillful JFM 2026 system (Spire) falls below ACC 0.5 by week 3. The Z500 skill horizon is correspondingly longer than the precipitation horizon, consistent with the well-established result that large-scale circulation is predictable further into the subseasonal range than the surface-sensible weather it forces (Vitart et al., 2017; Rasp et al., 2024).

7 Limitations

We list the limitations of this study candidly, as several materially bound how the results should be interpreted.

Two seasons, not a multi-year hindcast. JFM 2026 and JJAS 2019 are single, specific seasons rather than a multi-year hindcast archive. The two-season contrast isolates a real regime difference (Section 6), but neither season's result should be read as a long-run climatological skill estimate for that season type; both winter and monsoon predictability vary year to year with the state of ENSO and other low-frequency drivers (Mariotti et al., 2020; Domeisen et al., 2022), which this single-year comparison cannot sample.

Different precipitation truth in each season. JFM 2026 precipitation is verified against ERA5 daily totals because the IMD gridded product for 2026 is not yet available; JJAS 2019 precipitation is verified against IMD (Section 2.2). This is the principled choice given current data availability, but it means the *within-season* model rankings reported here are robust while the *absolute* precipitation skill level should not be compared directly across the two seasons, since ERA5 precipitation has its own documented regional biases relative to gauge-based observations (Hersbach et al., 2020).

Spire is absent from JJAS 2019. Spire AI-S2S forecasts were not available for the 2019 season, so the central winter finding—that an ML system can lead an operational benchmark over India—cannot itself be tested against the monsoon. The JJAS 2019 results show that the operational systems and FuXi-S2S do not retain useful monsoon rainfall skill beyond weeks 2–3, and the smaller DLESyM-inclusive subset shows no hidden long-lead circulation advantage for DLESyM, but we cannot directly test whether Spire specifically would retain a monsoon advantage.

Common climatology, no per-model bias correction. All anomalies use a single climatology per truth source (Section 4.2) rather than each system's own reforecast-derived climatology, so lead-dependent model drift is not removed before scoring. This is the standard approach for a cross-system comparison without access to every system's reforecast archive, but it means some of the long-lead skill loss documented in Section 5 may reflect uncorrected mean-state drift rather than loss of anomaly information per se.

Unequal and season-dependent ensemble sizes. Ensemble size ranges from 4 (UKMO) to 101 (ECMWF) members across the dynamical and ML-ensemble systems (Table 1), and Spire delivers a mean-and-spread product rather than individual members. UKMO's small ensemble in particular limits the precision of its probabilistic scores (Section 5.5) independent of the underlying forecast quality. More importantly, ECMWF and FuXi-S2S were archived with materially different ensemble sizes in the two seasons evaluated here (ECMWF: 101 members in JFM 2026, 51 in JJAS 2019; FuXi-S2S: 11 members in JFM 2026, 50 in JJAS 2019; Sections 3.2 and 3.4), reflecting what was available in each season's archive rather than a controlled experimental design. Probabilistic scores (CRPS, spread–skill ratio) for these two systems are consequently not built from a constant ensemble configuration across seasons, and any apparent change in their calibration between JFM and JJAS should be interpreted with this confound in mind.

Spire's CRPS uses a Gaussian convention. Because Spire is delivered only as a mean and standard deviation, it alone is scored with the closed-form Gaussian CRPS, whereas the member-based systems use the empirical estimator (Section 4.3). We quantified the effect of this asymmetry by scoring the member-based systems both ways: for Z500 it is negligible (median difference under 1%, at most 2.7%), so Z500 CRPSS comparisons are robust, but for skewed precipitation the Gaussian score is systematically higher (≈20% in the mean, growing with lead). The Gaussian convention therefore penalizes Spire's precipitation CRPSS relative to the member-based systems, so ECMWF's week-2–5 precipitation-CRPSS lead over Spire (Section 5.5) is in part a scoring-convention effect rather than a pure skill difference; the deterministic-versus-probabilistic contrast we draw (Section 6) is therefore stated most confidently for Z500, where the convention is immaterial. Closing this gap fully would require access to Spire's individual members, which were not provided.

Significance is bootstrap-based, not analytic, and conditions on one season. Reported skill differences are accompanied by a paired bootstrap over initialization dates (Section 4.5), which is what underwrites the significance language used in Section 5: it establishes, for example, that Spire's precipitation-ACC lead over ECMWF is significant at every lead, while its lead over FuXi-S2S and UKMO is significant only through week 3 and is within sampling uncertainty at weeks 4–6 (the lead over NCEP remains significant at weeks 4 and 6 but not week 5). Two caveats bound this. First, the bootstrap resamples *initializations within a season*; it therefore quantifies robustness to which dates happened to fall in this winter or this monsoon, but it does *not* and cannot represent the larger uncertainty of having sampled only a single season of each type—so a difference flagged as significant here is significant *for JFM 2026 (or JJAS 2019)*, not established as a climatological property of the season type. Second, we report per-comparison intervals without a multiple-comparisons correction across the full set of model pairs, lead weeks, and variables; individual borderline results (a 95% interval that barely excludes zero) should be read with that in mind, though the central findings—Spire's winter precipitation lead over ECMWF and its Z500 advantage over the other ML systems, and the monsoon skill collapse by week 3—rest on differences far from the significance boundary.

JFM 2026 truth source selection. For JFM 2026, the ERA5 truth field used for a given variable and verification window is selected automatically from among several candidate ERA5 archive files (a primary JFM-specific daily file and one or more fallback daily archives) based on which candidate has the most complete day coverage for that window; the file actually used is recorded per row in the `truth_source` column of the result data but is not summarized in this paper. In the published runs this fallback resolves to a single consistent source for the great majority of initializations, but a reader reproducing this analysis from a different or incomplete local archive state could in principle select a different candidate file for some windows and obtain marginally different truth values at the margins of the JFM 2026 record.

Fixed initialization sample across lead weeks. An earlier version of the JFM 2026 verification pipeline scored fewer initializations at the longest lead weeks, because the ERA5 truth record available at that time ended before some late-March initializations could be verified out to day 42; this risked mixing genuine skill decay with a shrinking, systematically later-weighted sample at long lead. The JFM 2026 results reported in this paper use a corrected run with the ERA5 truth record extended so that the full 90-initialization sample is scored identically at every lead week (verified directly against the result data), so the week-by-week ACC trajectories in Section 5.2 reflect skill decay on a fixed cohort rather than a changing sample composition.

Variable coverage: precipitation and Z500 only. This study scores precipitation and Z500 and does not report a 2-m temperature (T2M) comparison. T2M is the most data-limited variable in the available archives: UKMO and ECMWF 2-m temperature files contain only a single forecast step per initialization (a download-time error rather than a true data gap; Section 3.5), NCEP carries no T2M-equivalent channel, and although Spire, FuXi-S2S, and DLESyM do provide T2M, a verified daily-mean reference and valid-time convention were not established for these fields in time for this analysis. Rather than report a T2M comparison we could not yet fully verify, we restrict the paper to the two variables (precipitation and Z500) for which the truth, climatology, and valid-time conventions are confirmed. DLESyM additionally does not provide a precipitation channel (Section 3.3). A verified T2M comparison is a natural extension once these reference issues are closed.

8 Conclusions

We presented a two-case-study subseasonal benchmark over India. The JFM 2026 winter case evaluates six S2S systems—three machine-learning systems (Spire AI-S2S, FuXi-S2S, DLESyM) and three operational dynamical systems (ECMWF, UKMO, NCEP)—for precipitation and Z500 on 90 daily initializations. The JJAS 2019 monsoon case evaluates ECMWF, UKMO, NCEP, and FuXi-S2S on 35 common Monday/Thursday initializations, with DLESyM treated as a smaller sensitivity comparison where its available variables permit. The main conclusions are as follows.

(1) Spire AI-S2S leads winter precipitation forecasting over India, and degrades most gracefully for Z500. It attains the highest precipitation ACC and lowest RMSE at every lead week (week-1 ACC 0.78 versus 0.73 for ECMWF/UKMO); a paired bootstrap confirms this lead over ECMWF is significant at every lead and over all precipitation-capable systems through week 3; at weeks 4–6 it remains highest by point estimate but is statistically distinguishable only from ECMWF (and from NCEP at weeks 4 and 6). For Z500, no individual system—including Spire—holds skill flat through week 6, but Spire is the highest- or second-highest individual system at every lead from week 3 onward (ACC 0.39–0.66, versus ECMWF's fall to 0.13 by week 6 and FuXi-S2S/DLESyM's significant collapse into negative territory by week 4), making it the most consistently skillful system across the full six-week horizon without being immune to lead-time decay.

(2) The winter ML advantage should not be generalized to the monsoon. In the 35-date JJAS 2019 benchmark, every available precipitation system loses usable weekly rainfall skill by week 3–4, with no system retaining a meaningful deterministic edge at longer leads. The smaller DLESyM-inclusive Z500 sensitivity does not reveal a hidden monsoon long-lead ML advantage.

(3) The ML systems are not interchangeable. In JFM 2026, Spire, FuXi-S2S, and DLESyM look similar at week 1 but diverge sharply by week 3–4: FuXi-S2S and DLESyM fall to negative Z500 ACC by week 4, while Spire remains positive through week 6 even as its own skill declines. Long-lead stability is therefore a property that must be evaluated directly and cannot be inferred from short-lead skill alone.

(4) Deterministic and probabilistic rankings disagree. ECMWF leads JFM 2026 precipitation CRPSS from weeks 2–5 despite Spire's higher ACC throughout, while FuXi-S2S narrowly leads CRPSS at week 6. A system favored by deterministic verification is therefore not automatically the better choice for probabilistic, risk-based use.

Within the limits of two single seasons (Section 7), these results give an early, honest picture of where ML subseasonal forecasting adds value over India—the synoptically organized winter—and where the problem remains genuinely hard for the available ML and operational systems evaluated here—the summer monsoon. The verification framework, code, and result tables are archived to support the next necessary extensions: multi-year hindcasts, IMD verification for 2026 rainfall when available, access to Spire's individual ensemble members, and per-model reforecast climatologies.

Data and Code Availability

ERA5 reanalysis is available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu`; Hersbach et al. 2020); the Analysis-Ready, Cloud-Optimized subset used here follows Carver and Merose (2023). IMD gridded daily rainfall follows Pai et al. (2014) and is distributed by the India Meteorological Department. ECMWF extended-range, UKMO GloSea6, and NCEP CFSv2 forecasts are available via the ECMWF Copernicus Data Store under the terms of the S2S database (Vitart et al., 2017). FuXi-S2S model weights and inference code are publicly available on Zenodo and GitHub respectively. Spire AI-S2S forecasts were provided under a research data-use agreement with Spire Global. DLESyM forecasts were obtained from the model providers (Cresswell-Clay et al., 2025). All verification, analysis, table-generation, and figure-generation code for this study, including the scripts that regenerate the metric tables and lead-time figures directly from the underlying result files, is available at `https://github.com/Artamta/s2s-analysis`.

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A Supplementary Results

This appendix collects the supplementary regional, spatial, and scatter diagnostics referenced in Sections 5.2–5.7.

A.1 Supplementary regional and spatial diagnostics

Figure 9 is the week-1 JFM 2026 precipitation ACC scorecard by IMD region referenced in Section 5.7, shown here as a model-by-region heatmap. Figure 10 is the Z500 counterpart of the regional precipitation panels (Figure 6): regional Z500 skill is both higher and more uniform across regions than precipitation, and Spire's more graceful long-lead decay (Section 5.4) is visible in every region. Figure 11 maps the JFM 2026 weeks-1–6 mean precipitation bias per system; the shared dry bias over the Northeast Himalayan foothills (Section 5.7) is evident as a common brown feature, while UKMO and NCEP carry a broader wet tint over central and peninsular India.

A.2 Forecast-versus-observed scatter

Figures 12 and 13 show, for JFM 2026, the grid-level density scatter of forecast anomaly against observed (ERA5) anomaly, pooling all Indian land points and all initializations within each lead week (rows: weeks 1, 3, 6). A skilful forecast concentrates along the 1:1 line; the annotated pooled Pearson correlation r (distinct from the domain-mean, per-initialization ACC of the main text) contracts toward zero with lead. For precipitation (Figure 12) Spire has the tightest week-1 relationship ($r = 0.73$) and all systems degrade to $r \approx 0.1$–0.16 by week 6; for Z500 (Figure 13) the near-Gaussian anomalies scatter more symmetrically, and the FuXi-S2S and DLESyM collapse into weak/negative correlation by week 6 is visible as a rotation of the point cloud away from the 1:1 line.

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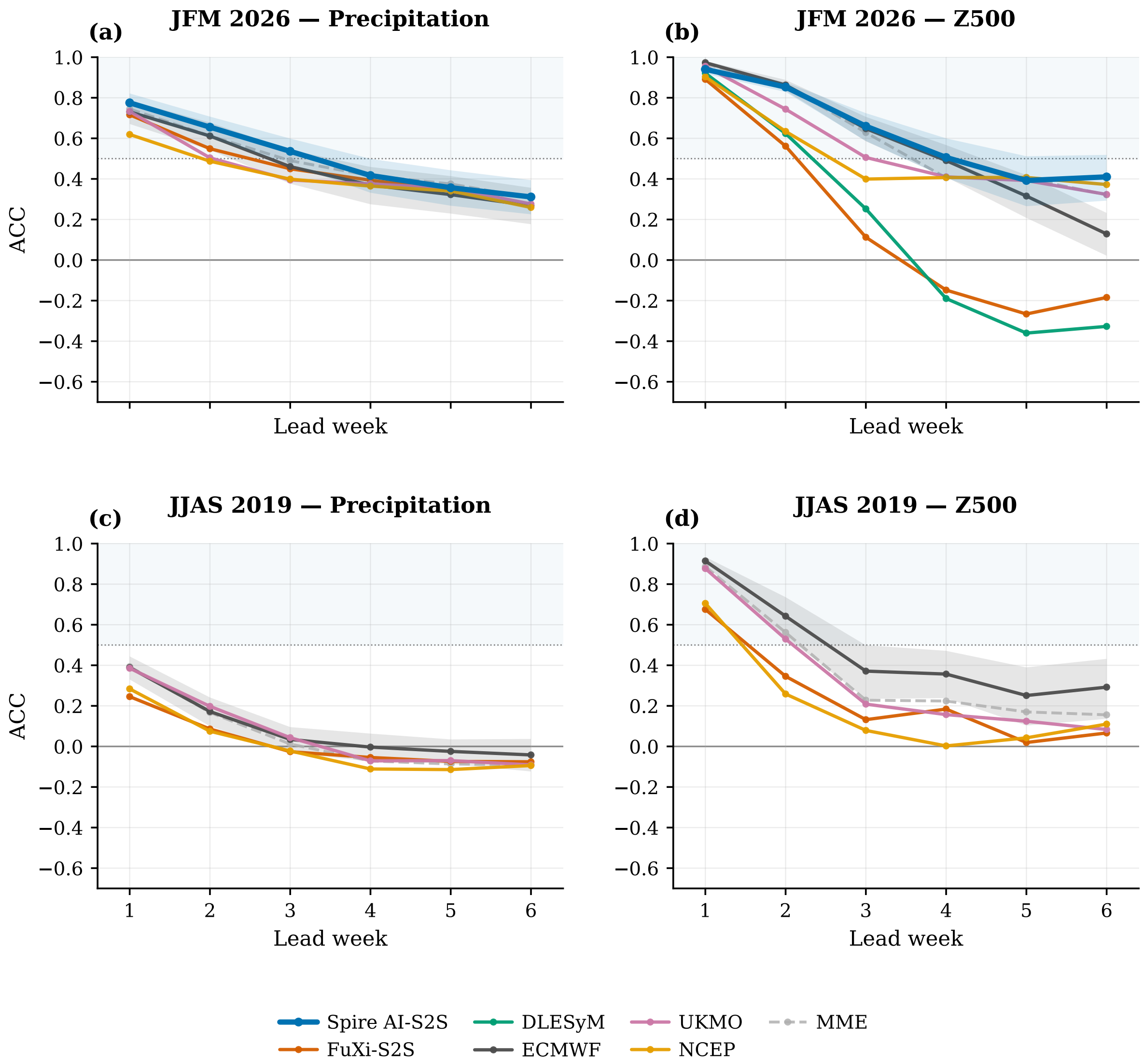

Figure 2: All-India anomaly correlation (ACC) versus lead week. Top row: JFM 2026 winter case with Spire AI-S2S, FuXi-S2S, DLESyM, ECMWF, UKMO, NCEP, and the MME where available. Bottom row: JJAS 2019 monsoon case on 35 common Monday/Thursday initializations for ECMWF, UKMO, NCEP, FuXi-S2S, and the MME. Columns: precipitation, Z500. Dotted line ACC$=0.5$; solid line ACC$=0$; light horizontal band marks ACC≥ 0.5. Shaded envelopes around the Spire (blue) and ECMWF (grey) curves are 95% paired-bootstrap confidence intervals over initializations (Section 4.5); bands for the other systems are omitted for legibility. DLESyM provides no precipitation and Spire is unavailable for JJAS.

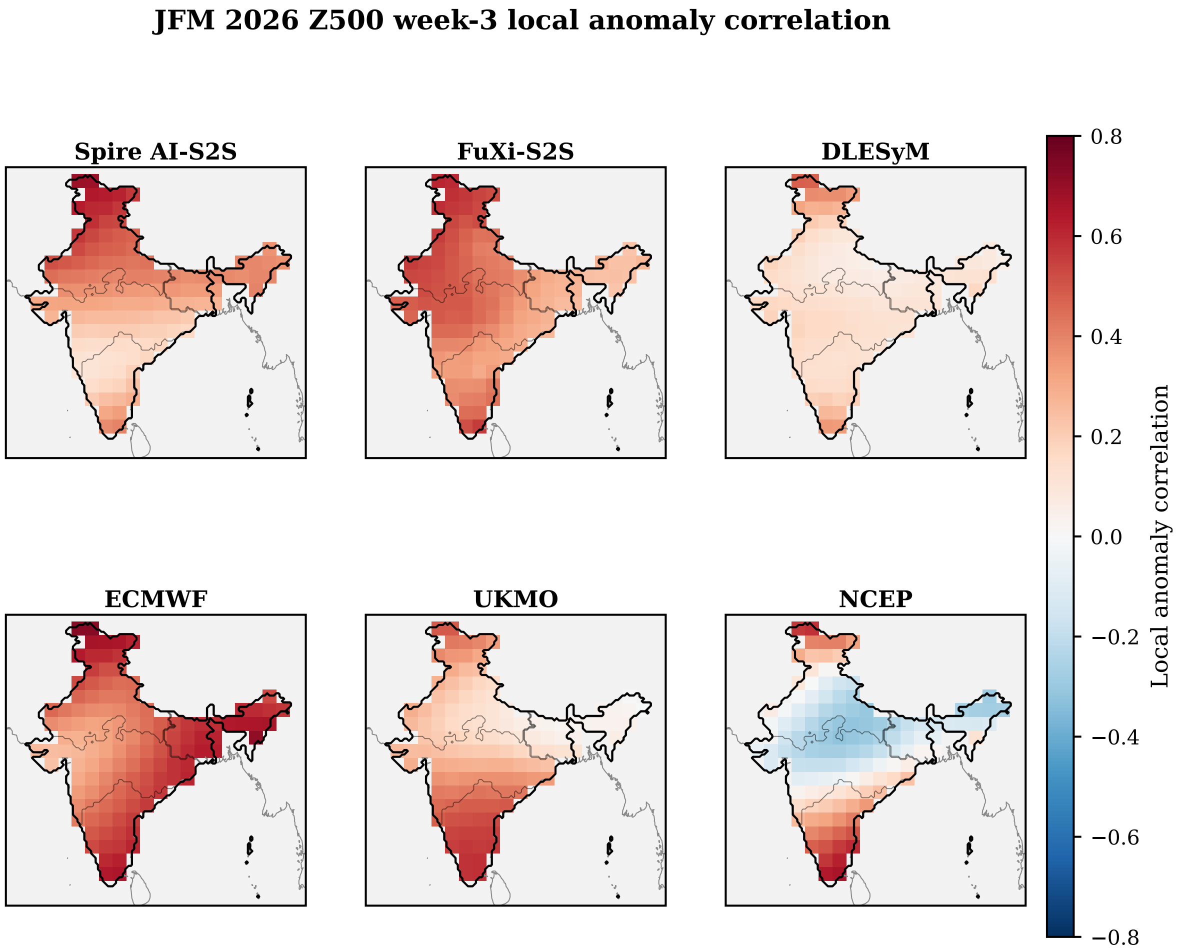

Figure 3: JFM 2026 week-3 Z500 local anomaly correlation—the temporal correlation, across the 90 initializations, between each system's anomaly and the ERA5 anomaly at every 1.5° land grid point. Red is positive (skillful), blue negative (anticorrelated). This per-cell diagnostic complements the domain-mean, per-initialization ACC reported in the tables (Section 4.3).

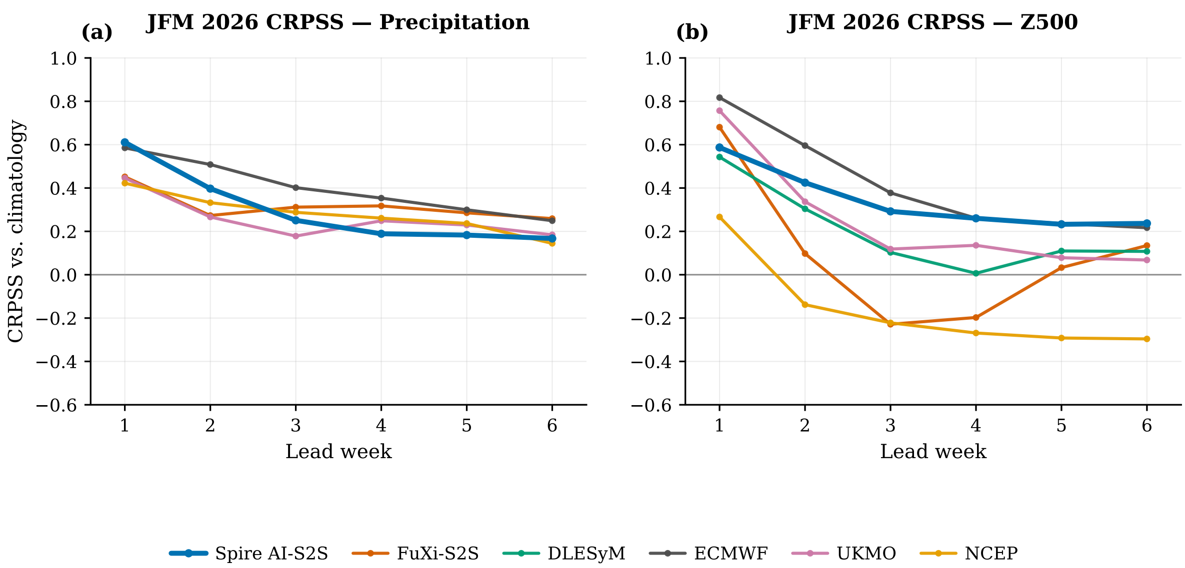

Figure 4: JFM 2026 all-India CRPSS versus lead week for precipitation (left) and Z500 (right). Positive values indicate probabilistic skill above climatology.

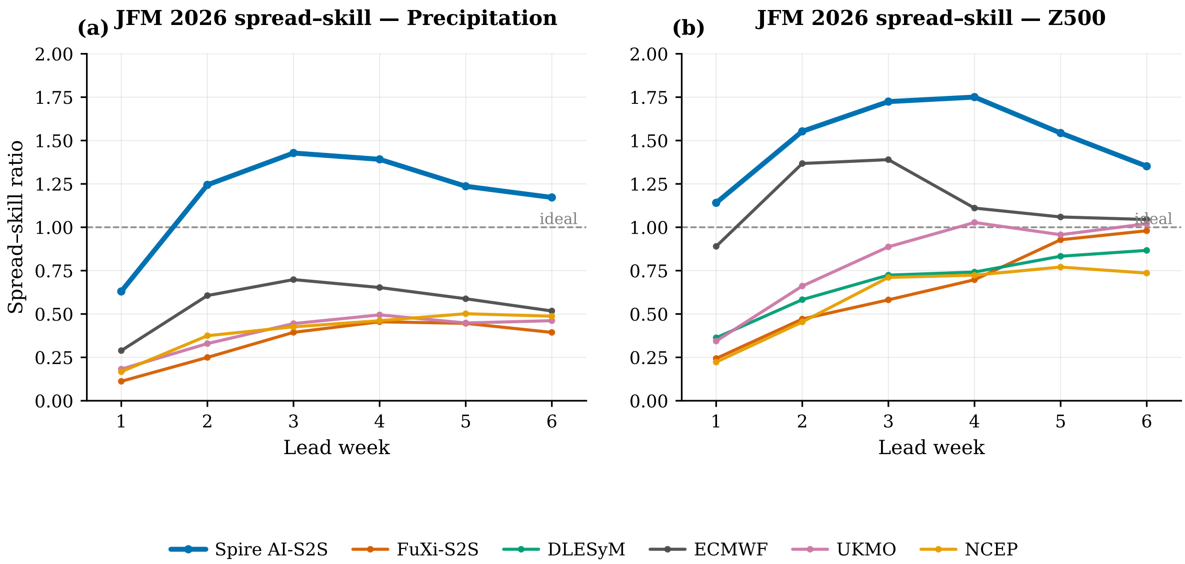

Figure 5: JFM 2026 spread–skill ratio versus lead week; a well-calibrated ensemble lies near unity (dashed line). Values < 1 indicate under-dispersion (overconfidence).

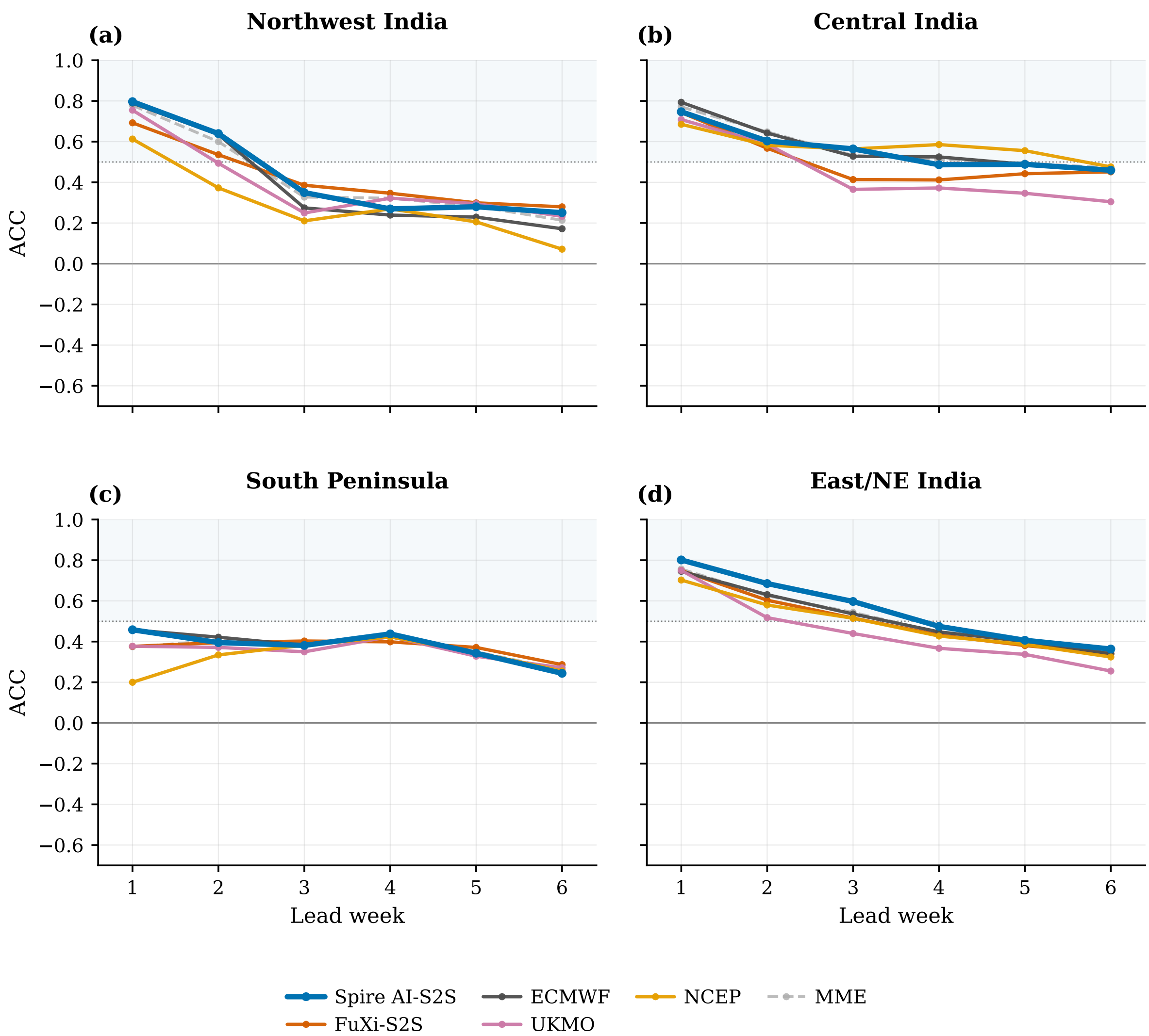

Figure 6: JFM 2026 precipitation ACC versus lead week for each of the four IMD homogeneous regions (Section 2.1), all systems. Dotted line ACC= 0.5; solid line ACC= 0; shaded band marks ACC≥ 0.5. DLESyM has no precipitation channel. The corresponding week-1 scorecard and the Z500 version are in Appendix A (Figures 9 and 10).

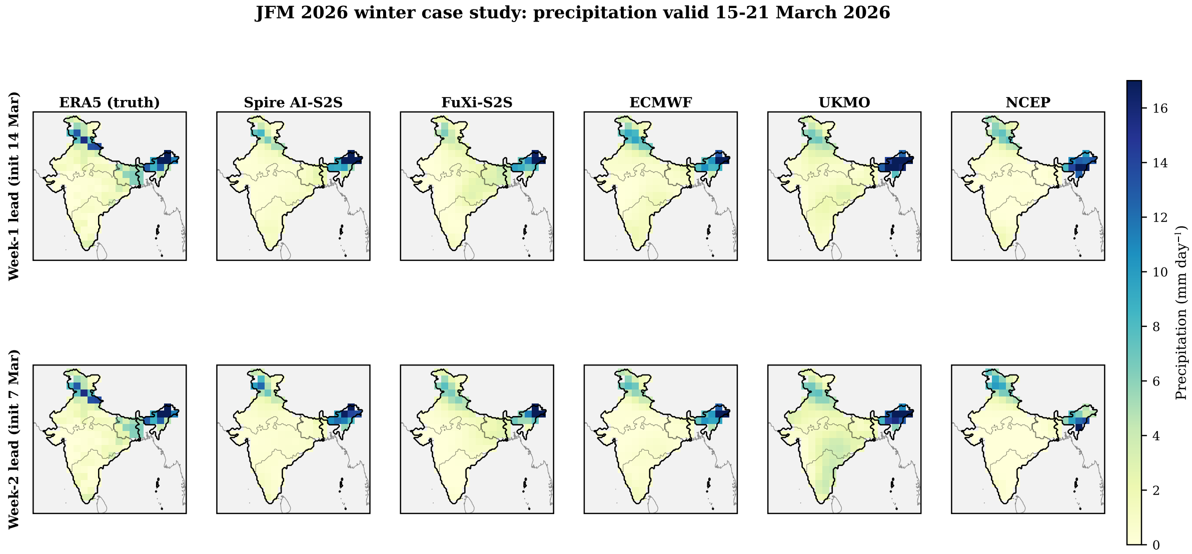

Figure 7: JFM 2026 winter rainfall case study: precipitation (mm day^{-1}) verifying 15–21 March 2026. Columns: ERA5 truth and each precipitation-capable system. Rows: week-1 lead (initialized 14 March) and week-2 lead (initialized 7 March). All panels share the common 1.5° verification grid and colour scale. DLESyM has no precipitation channel and Spire is included as its ensemble mean.

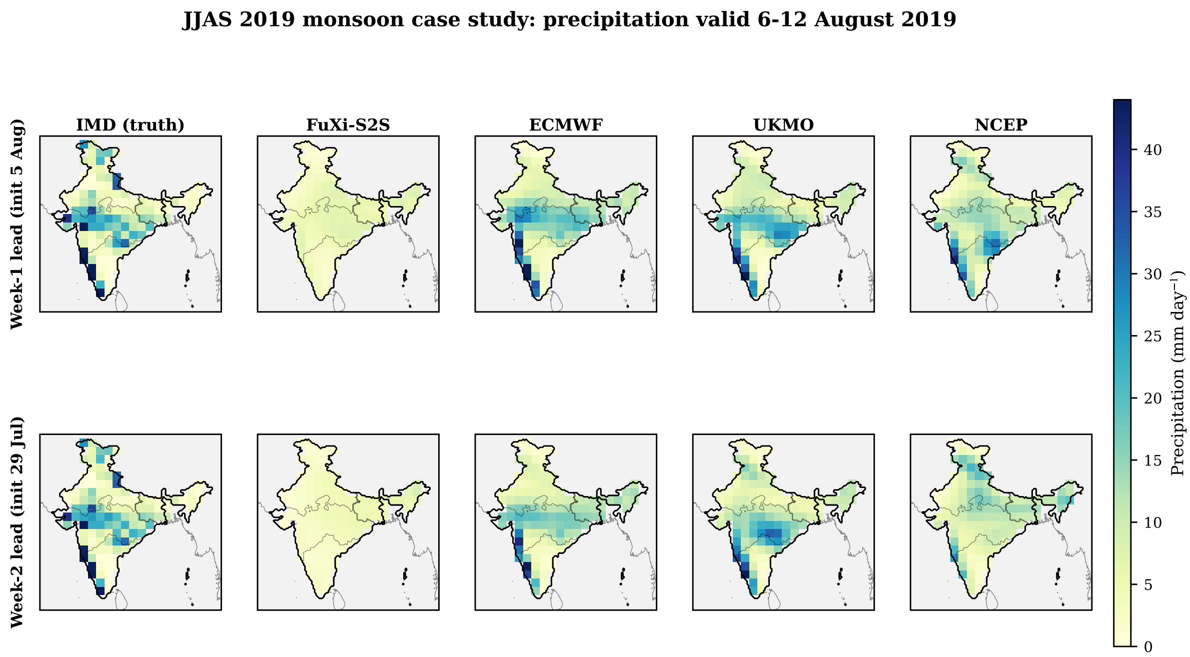

Figure 8: JJAS 2019 monsoon rainfall case study: precipitation (mm day^{-1}) verifying 6–12 August 2019. Columns: IMD gridded-rainfall truth and each precipitation-capable system available for JJAS 2019 (ECMWF, UKMO, NCEP, FuXi-S2S). Rows: week-1 lead (initialized 5 August) and week-2 lead (initialized 29 July). All panels share the common 1.5° verification grid and colour scale.

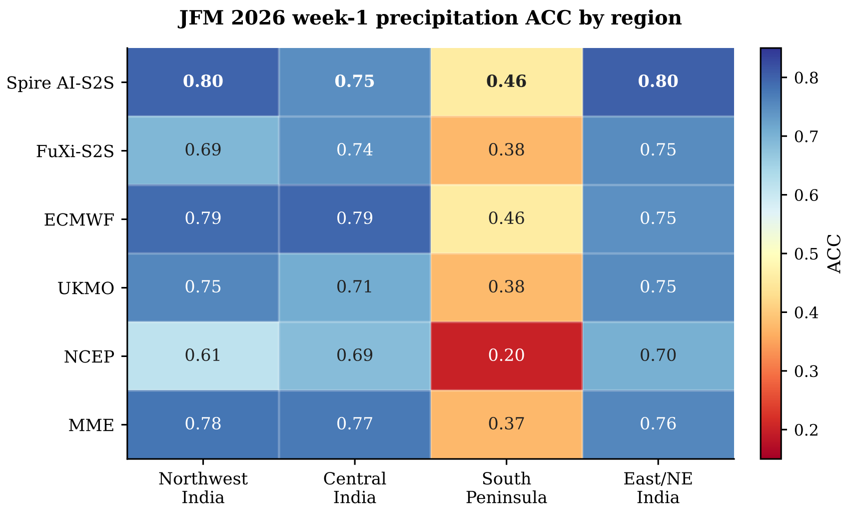

Figure 9: JFM 2026 week-1 precipitation ACC by IMD homogeneous region (model × region heatmap). This is the scorecard referenced from Section 5.7.

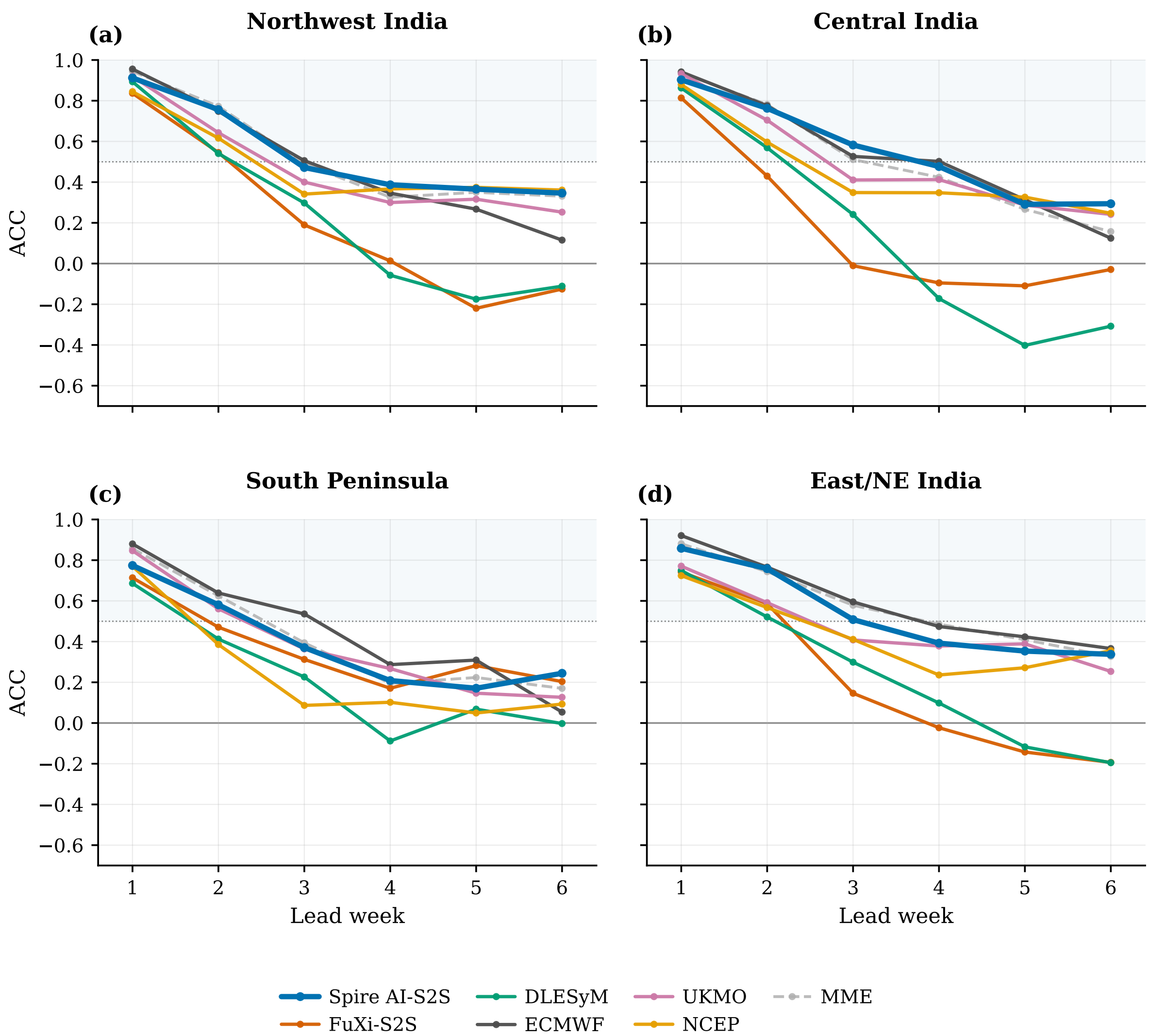

Figure 10: JFM 2026 Z500 ACC versus lead week for each of the four IMD homogeneous regions, all systems—the Z500 counterpart of Figure 6. Dotted line ACC= 0.5; solid line ACC= 0.

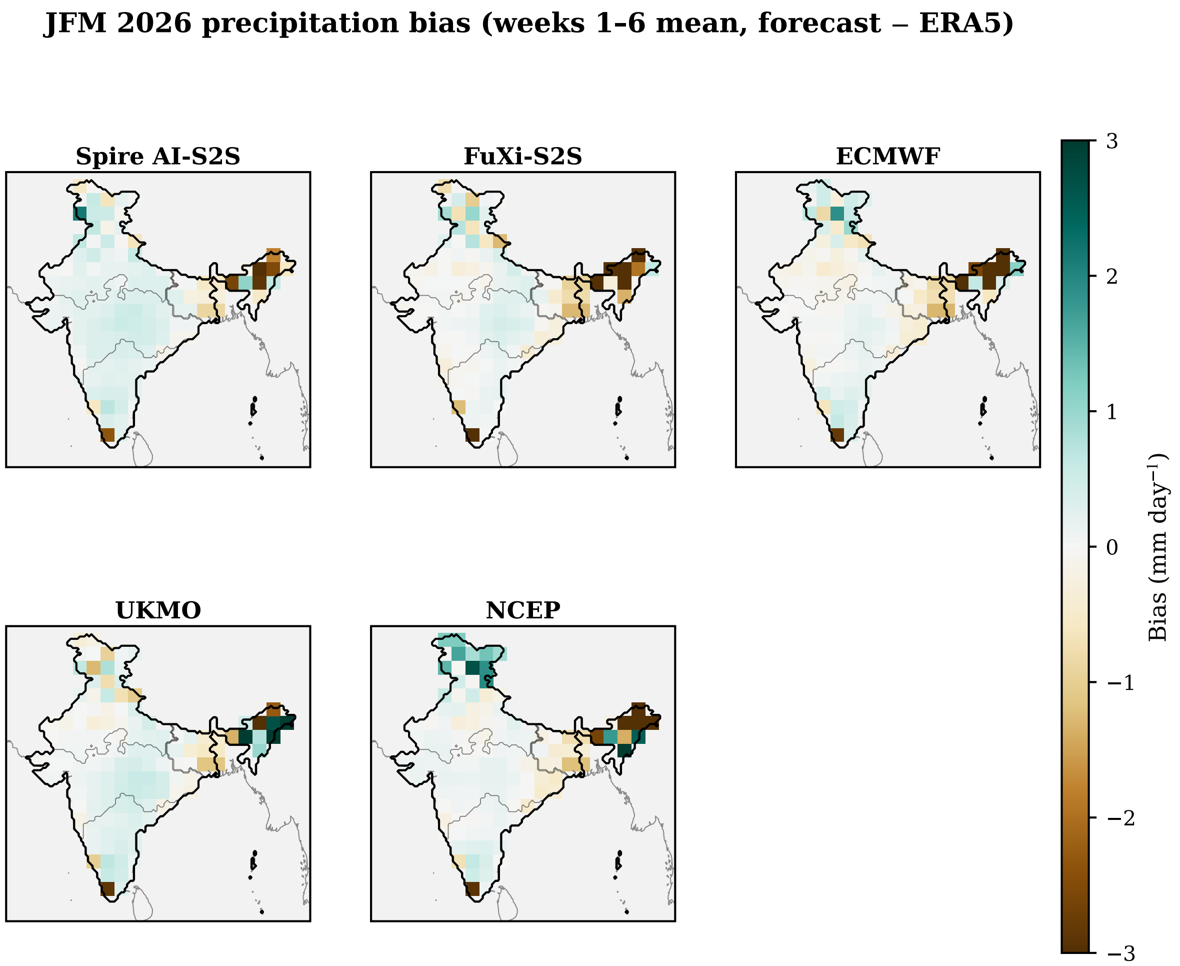

Figure 11: JFM 2026 precipitation bias (weeks-1–6 mean, forecast − ERA5) at every 1.5° land grid point, per system. Brown is dry, teal wet. DLESyM has no precipitation channel.

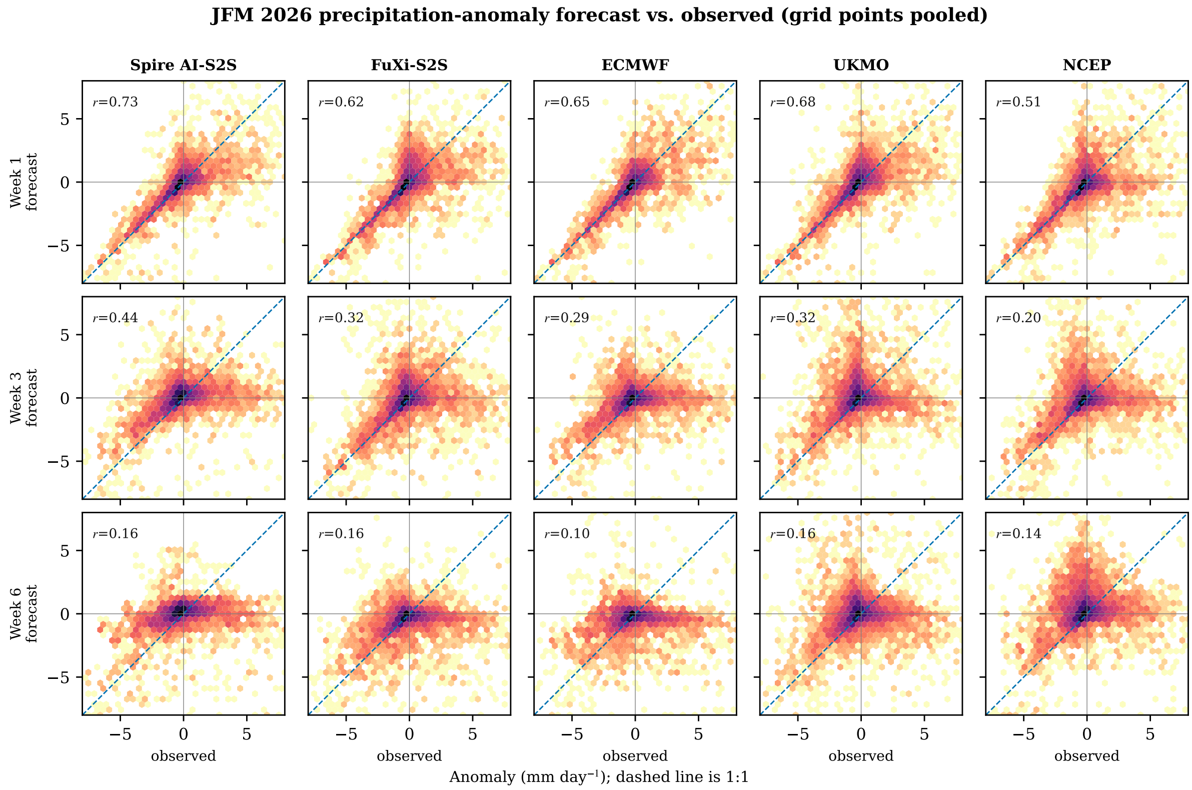

Figure 12: JFM 2026 precipitation-anomaly forecast versus observed (ERA5) density scatter, grid points and initializations pooled within each lead week. Rows: weeks 1, 3, 6; columns: systems. Dashed line 1:1; r is the pooled Pearson correlation.

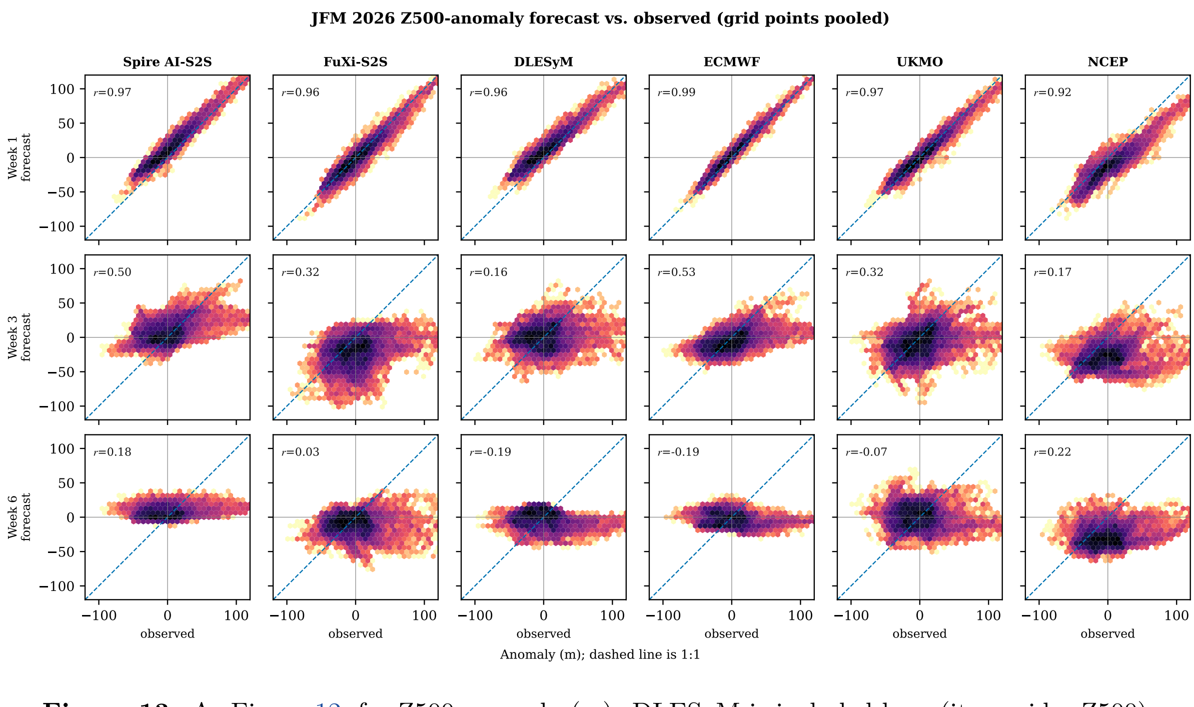

Figure 13: As Figure 12, for Z500 anomaly (m). DLESyM is included here (it provides Z500).